PathGA

April 23, 2018

In [47]:
```
import matplotlib.pyplot as plt
import numpy as np


num_movs = 150
dna_sz = num_movs*2
drct_bnd = [0, 1]
tx_cross = 0.8
tx_mutation = 0.0001
population = 100
epochs = 100
source = [10, 5]
destination = [0, 5]
wall = np.array([[5, 2], [5, 8]])

class GA(object):
    def __init__(self, dna_size, dna_bound, cross_rate, mutation_rate, pop_size, ):
        self.dna_size = dna_size
        dna_bound[1] += 1
        self.dna_bound = dna_bound
        self.cross_rate = cross_rate
        self.mutate_rate = mutation_rate
        self.pop_size = pop_size

        self.pop = np.random.randint(*dna_bound, size=(pop_size, dna_size))

    def DNA2product(self, dna, n_moves, start):
        pop = (dna - 0.5) / 2
        pop[:, 0], pop[:, n_moves] = start[0], start[1]
        lines_x = np.cumsum(pop[:, :n_moves], axis=1)
        lines_y = np.cumsum(pop[:, n_moves:], axis=1)
        return lines_x, lines_y

    def get_fitness(self, lines_x, lines_y, destination_pt, wall_line):
        distance = np.sqrt((destination_pt[0] - lines_x[:, -1]) ** 2 + (destination_pt[
        fitness = np.power(1 / (distance + 1), 2)
        points = (lines_x > wall_line[0, 0] - 0.5) & (lines_x < wall_line[1, 0] + 0.5)
        y_values = np.where(points, lines_y, np.zeros_like(lines_y) - 100)
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            bad_lines = ((y_values > wall_line[0, 1]) & (y_values < wall_line[1, 1])).max(a
            fitness[bad_lines] = 1e-6
            return fitness

        def select(self, fitness):
            idx = np.random.choice(np.arange(self.pop_size), size=self.pop_size, replace=Tr
            return self.pop[idx]

        def crossover(self, parent, pop):
            if np.random.rand() < self.cross_rate:
                i_ = np.random.randint(0, self.pop_size, size=1)
                cross_points = np.random.randint(0, 2, self.dna_size).astype(np.bool)
                parent[cross_points] = pop[i_, cross_points]
            return parent

        def mutate(self, child):
            for point in range(self.dna_size):
                if np.random.rand() < self.mutate_rate:
                    child[point] = np.random.randint(*self.dna_bound)
            return child

        def evolve(self, fitness):
            pop = self.select(fitness)
            pop_copy = pop.copy()
            for parent in pop:
                child = self.crossover(parent, pop_copy)
                child = self.mutate(child)
                parent[:] = child
            self.pop = pop


    class Line(object):
        def __init__(self, n_moves, goal_point, start_point, obstacle_line):
            self.n_moves = n_moves
            self.goal_point = goal_point
            self.start_point = start_point
            self.obstacle_line = obstacle_line

            plt.ion()

        def plotting(self, lines_x, lines_y):
            plt.cla()
            plt.scatter(*self.goal_point, s=200, c='r')
            plt.scatter(*self.start_point, s=100, c='b')
            plt.plot(self.obstacle_line[:, 0], self.obstacle_line[:, 1], lw=3, c='k')
            plt.plot(lines_x.T, lines_y.T, c='k')
            plt.xlim((-5, 15))
            plt.ylim((-5, 15))
```

```
            plt.pause(0.01)

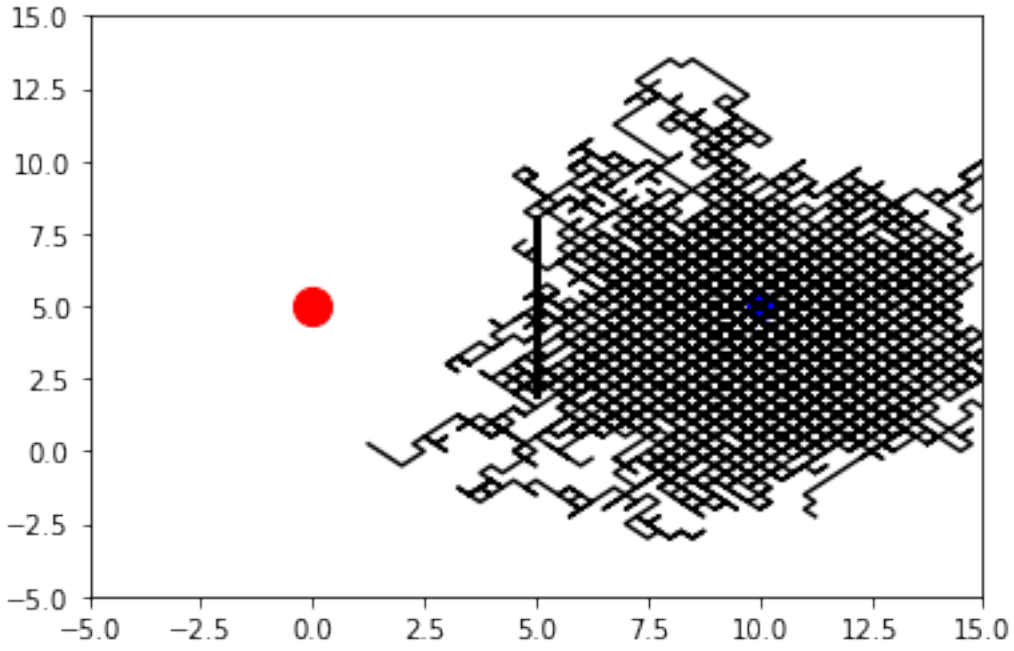

    ga = GA(dna_size=dna_sz, dna_bound=drct_bnd,
            cross_rate=tx_cross, mutation_rate=tx_mutation, pop_size=population)

    env = Line(num_movs, destination, source, wall)

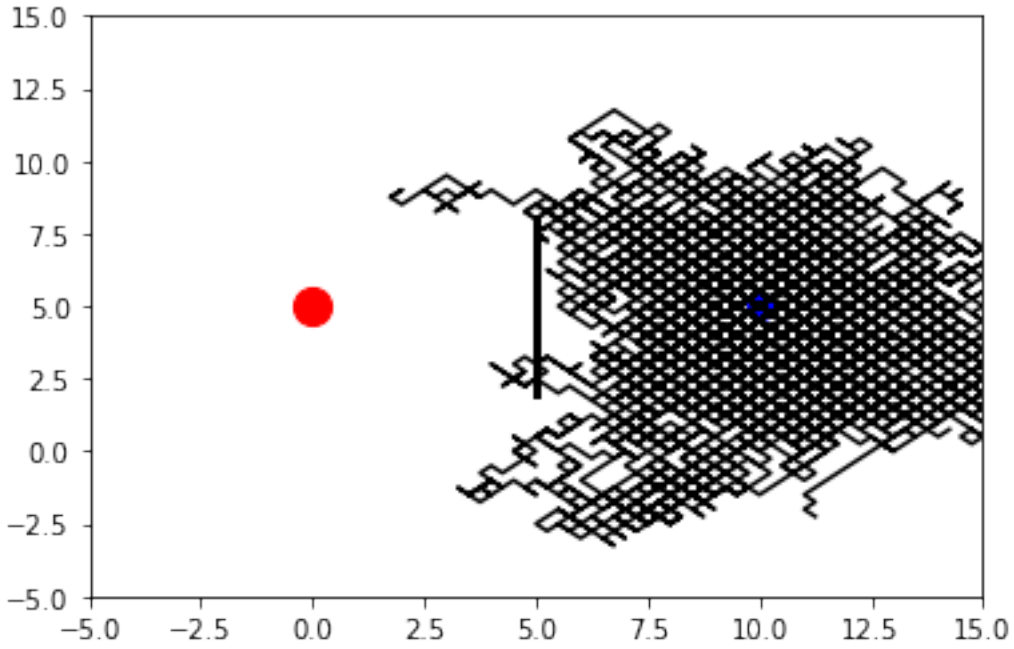

    for epoch in range(epochs):
        lx, ly = ga.DNA2product(ga.pop, num_movs, source)
        fitness = ga.get_fitness(lx, ly, destination, wall)
        ga.evolve(fitness)
        print('Ger:', epoch, '| best fit:', fitness.max())
        env.plotting(lx, ly)

        plt.ioff()
        plt.show()
```

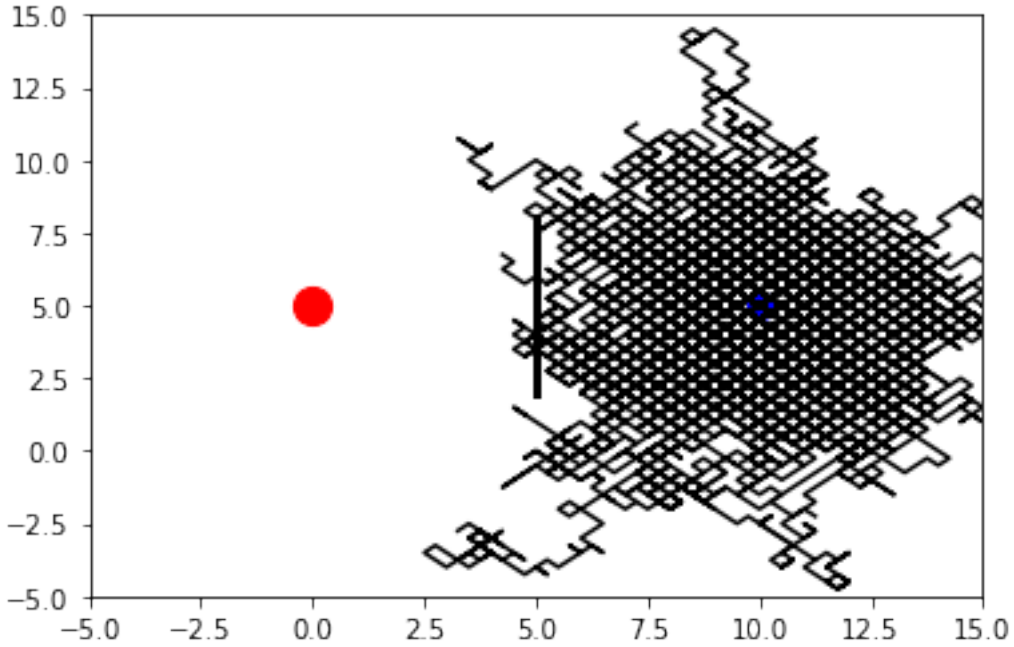
```
Ger: 0 | best fit: 0.0194272500736
```



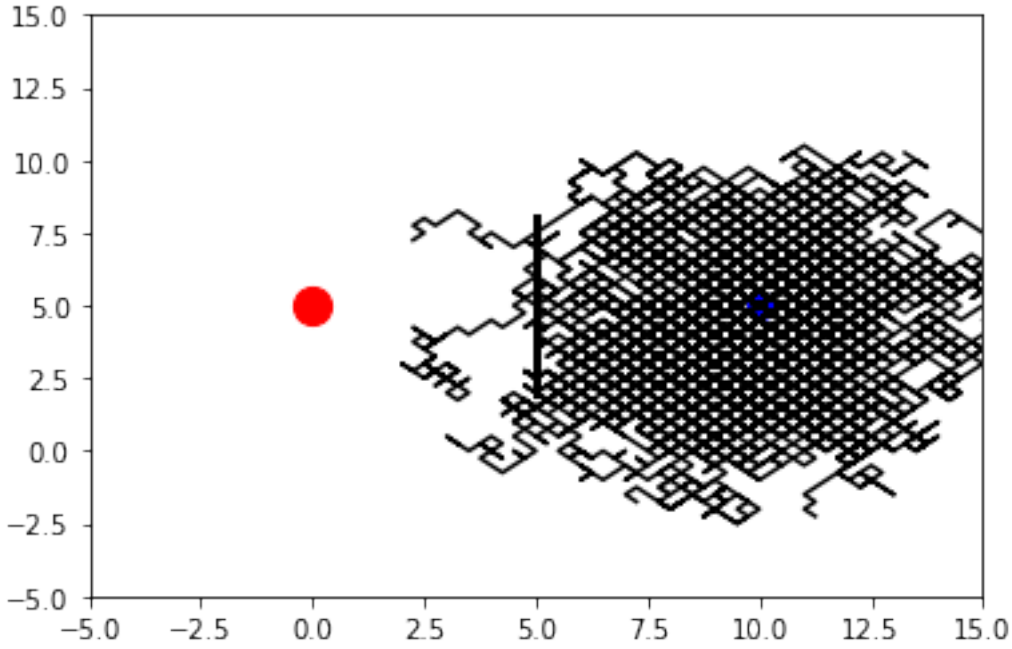

```
Ger: 1 | best fit: 0.0194272500736
```



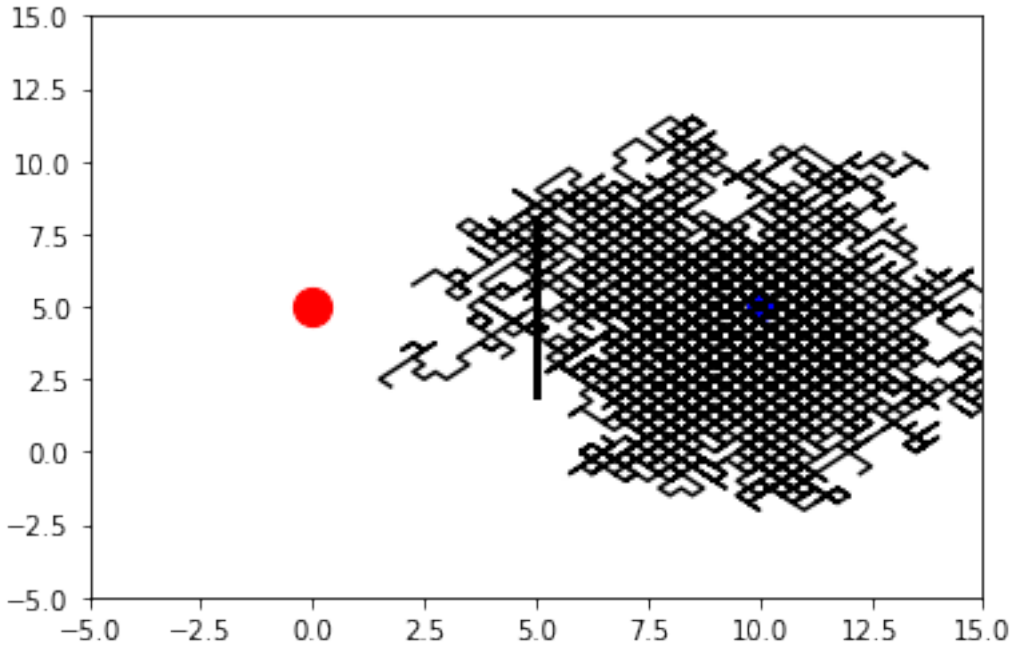

```
Ger: 2 | best fit: 0.0210998943224
```



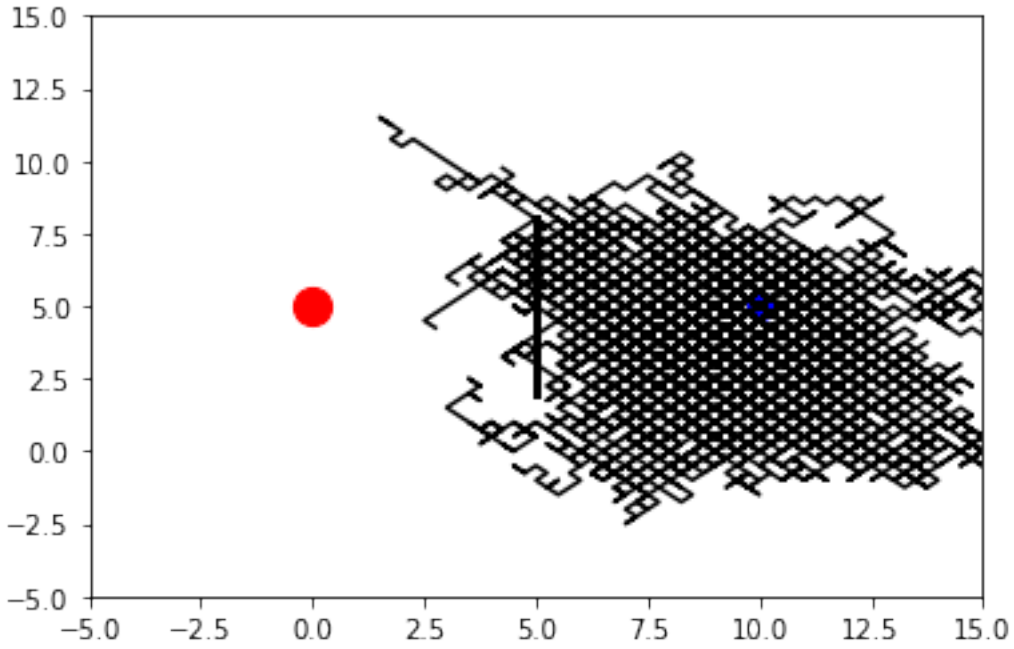
```
Ger: 3 | best fit: 0.0210998943224
```



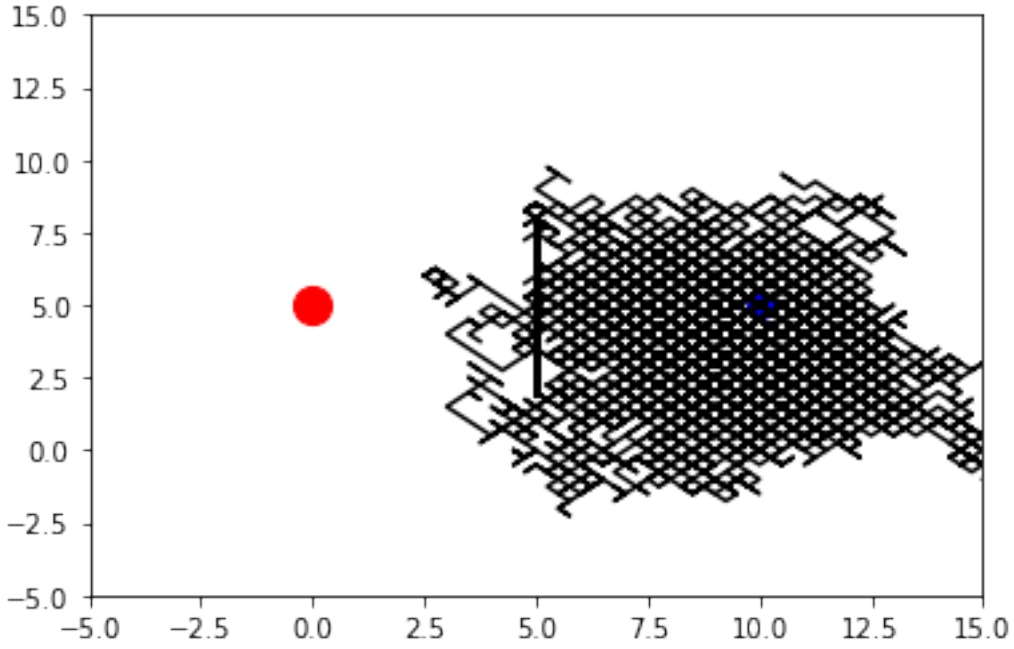

```
Ger: 4 | best fit: 0.0219125902932
```



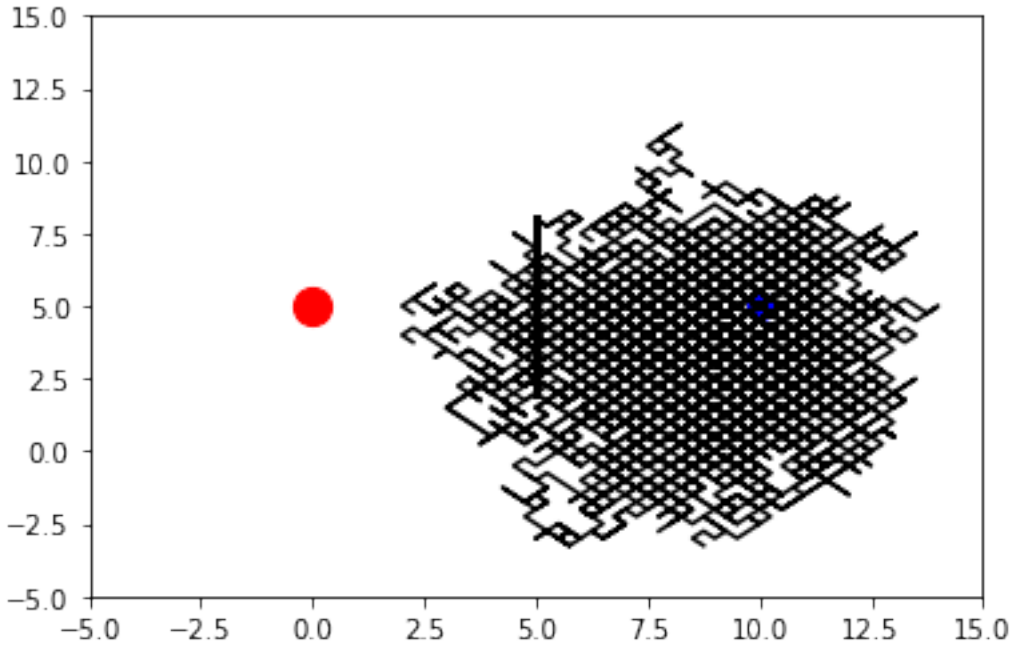
```
Ger: 5 | best fit: 0.0313228867729
```



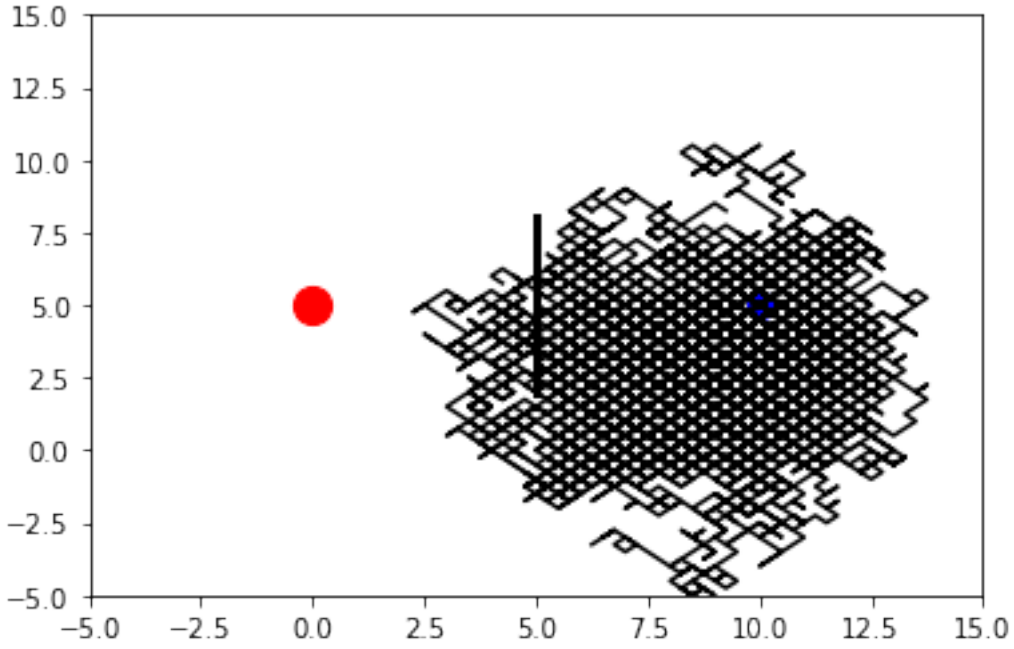

```
Ger: 6 | best fit: 0.0346363195289
```



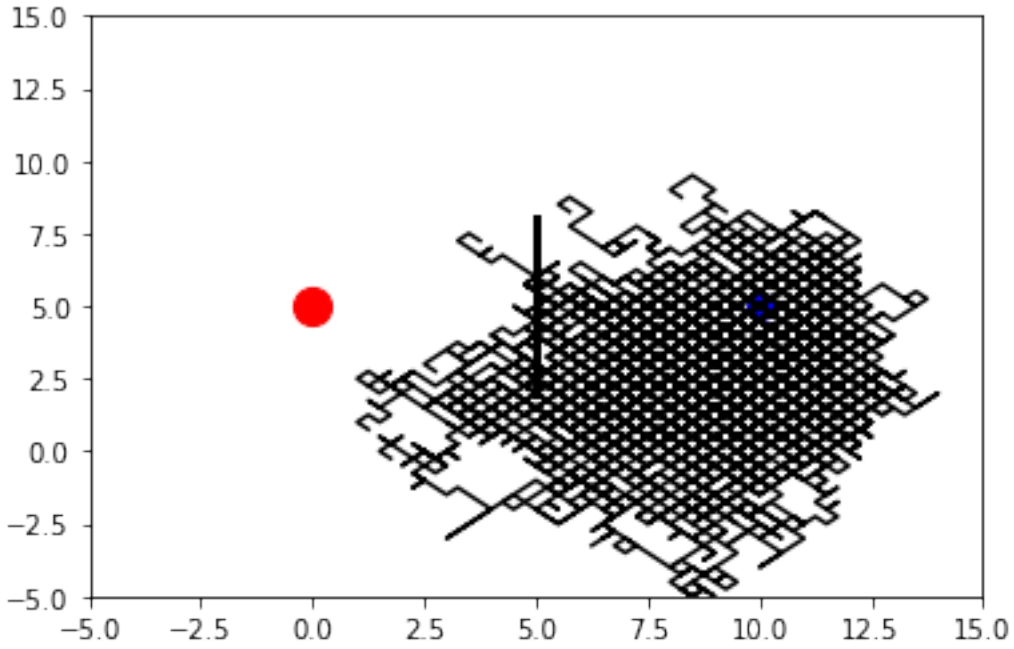
```
Ger: 7 | best fit: 0.0346363195289
```



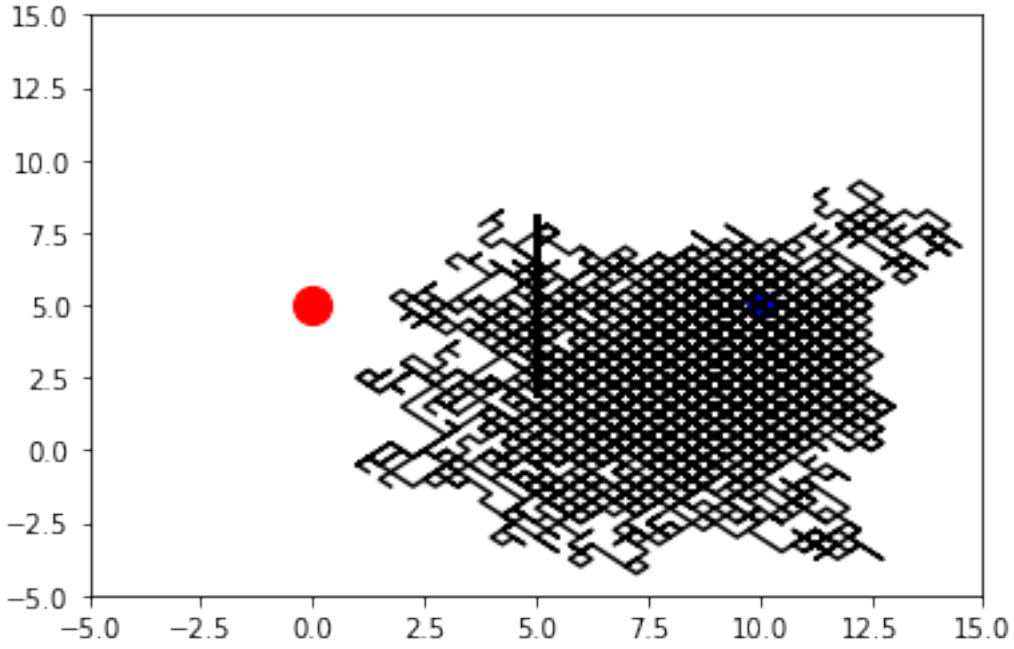

```
Ger: 8 | best fit: 0.0313228867729
```



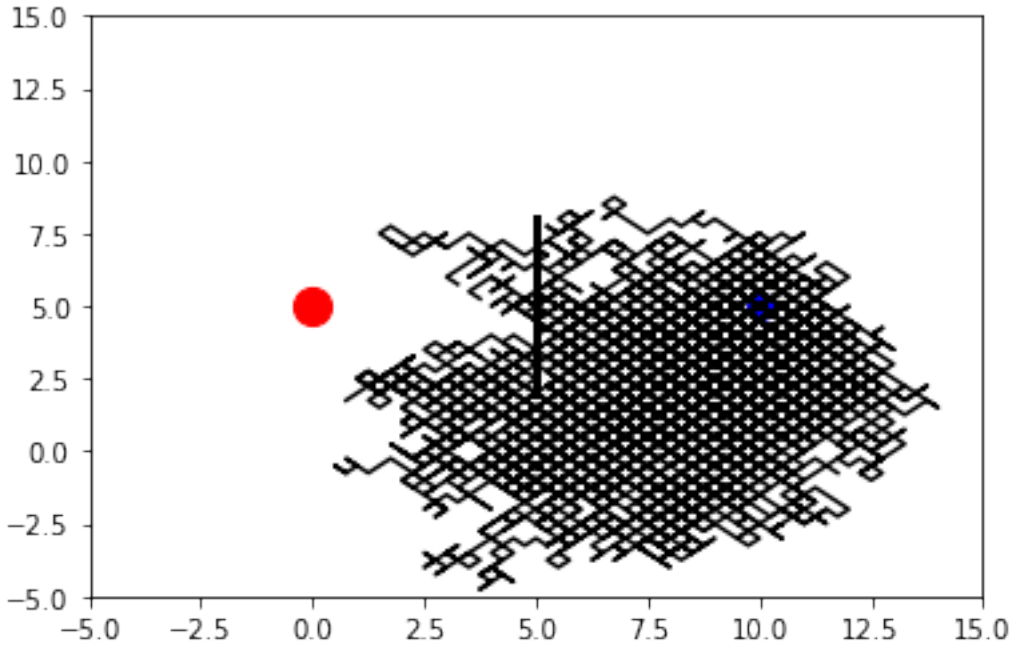
```
Ger: 9 | best fit: 0.061856219845
```



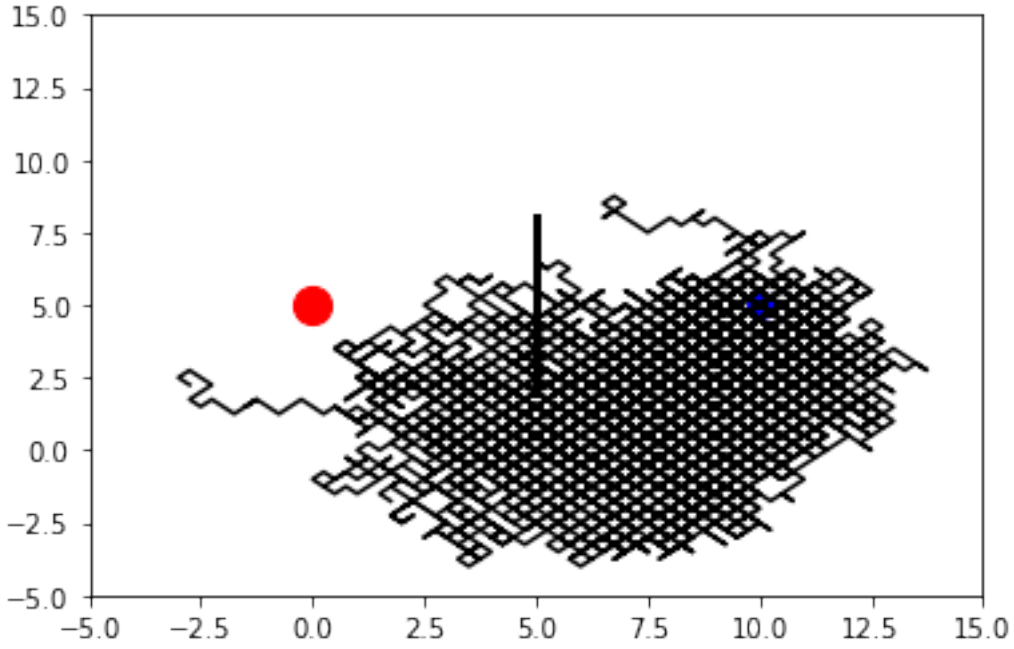

```
Ger: 10 | best fit: 0.061856219845
```



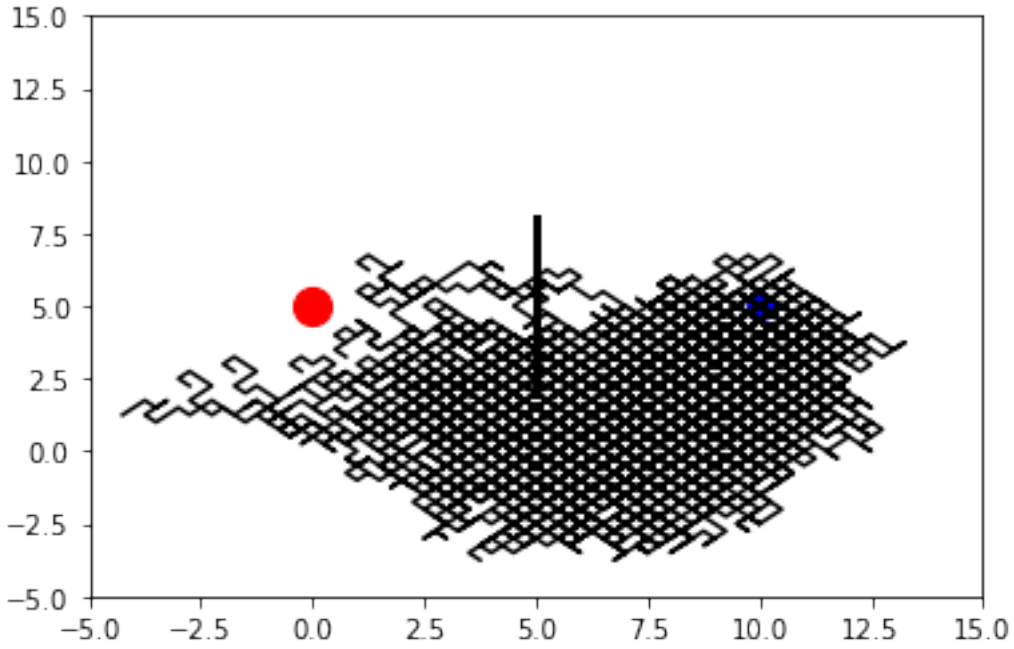
```
Ger: 11 | best fit: 0.061856219845
```



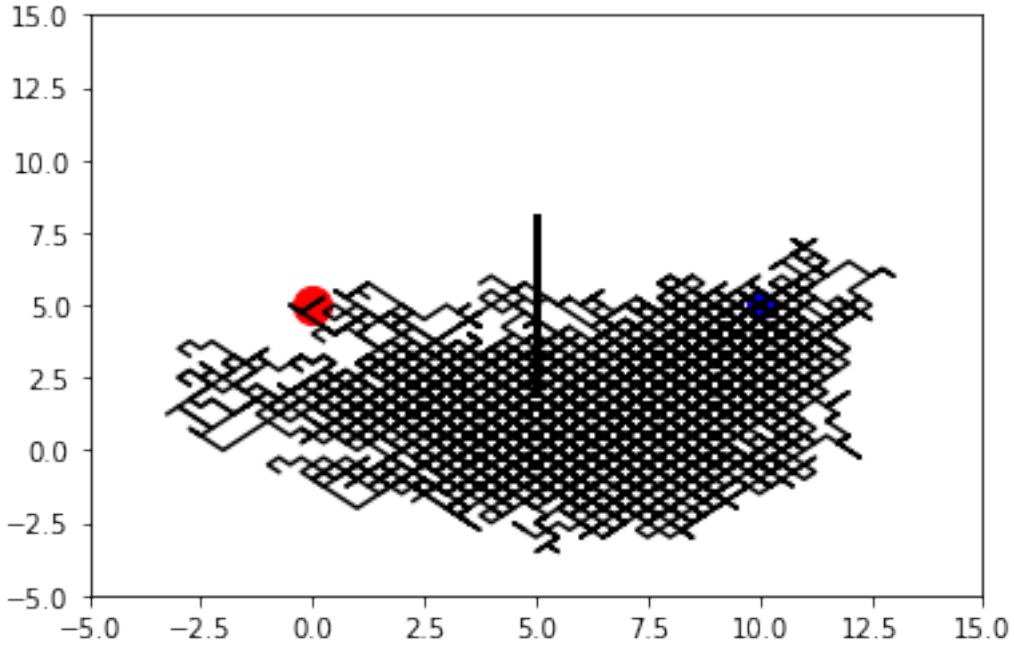

```
Ger: 12 | best fit: 0.061856219845
```



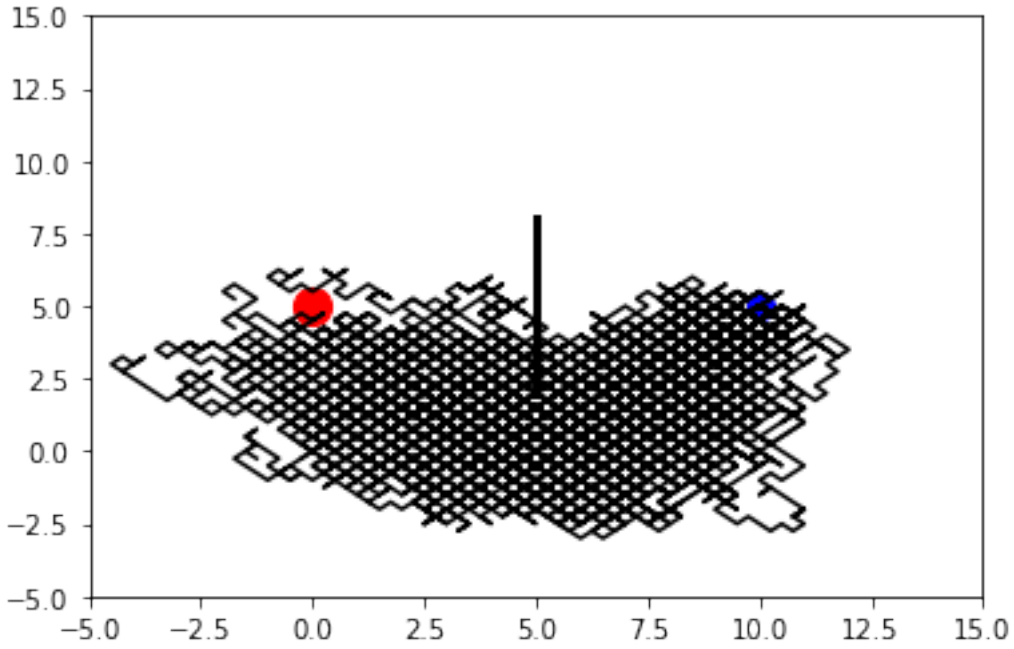
```
Ger: 13 | best fit: 0.0938729803302
```



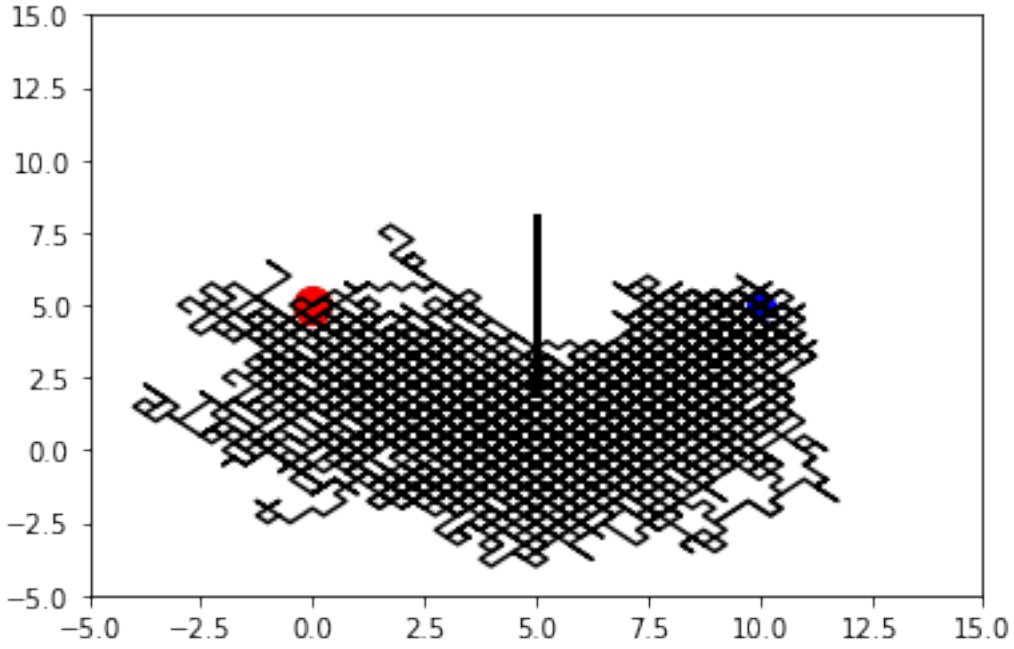

```
Ger: 14 | best fit: 0.193255022601
```



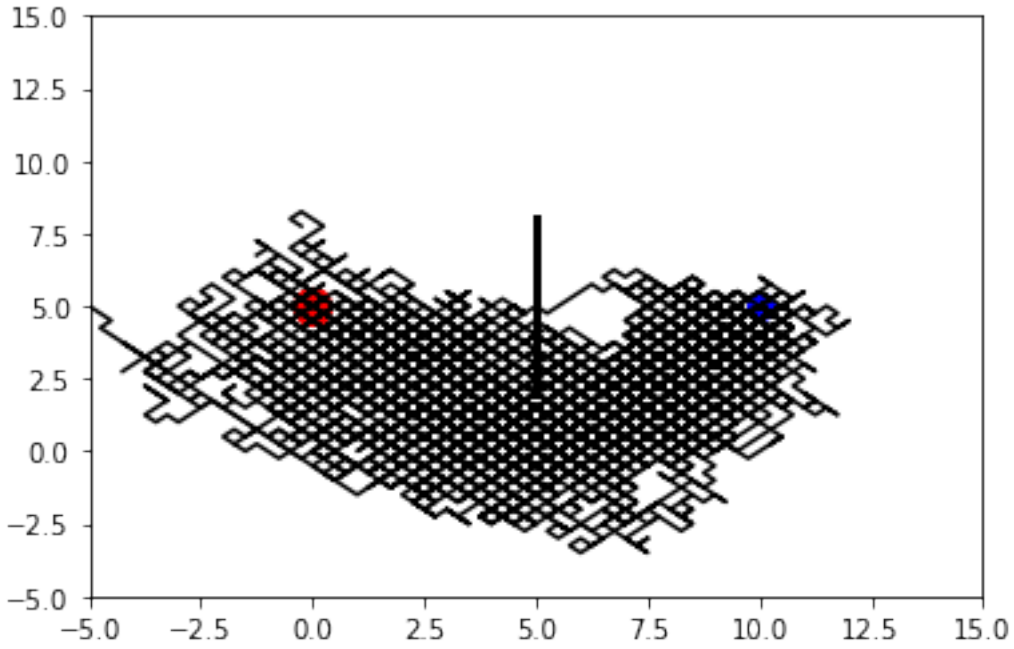
```
Ger: 15 | best fit: 0.193255022601
```



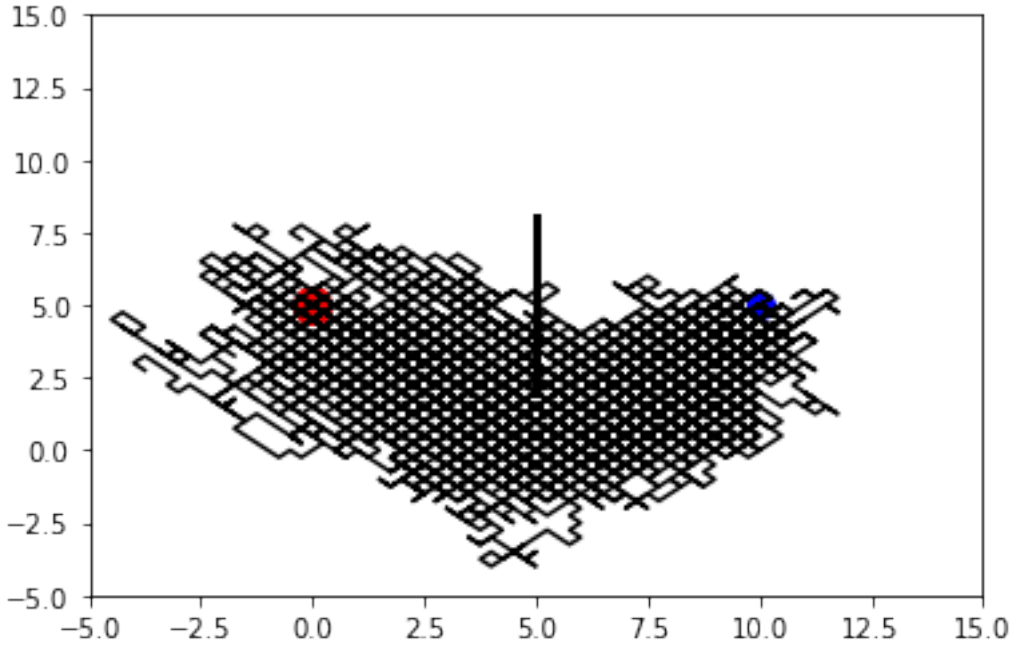

```
Ger: 16 | best fit: 0.545819714369
```



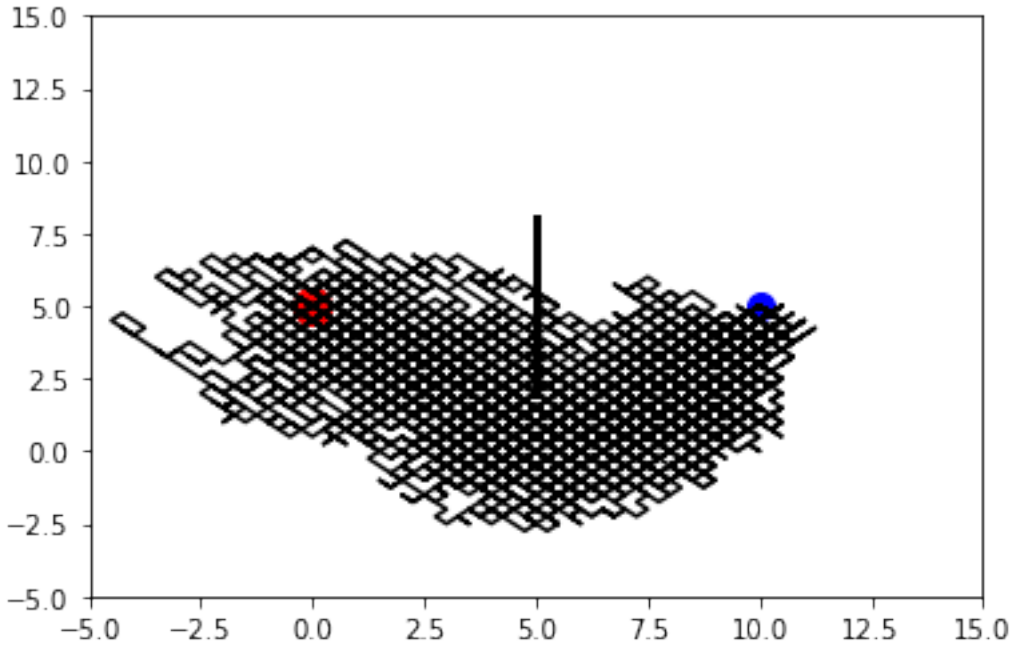
```
Ger: 17 | best fit: 0.545819714369
```



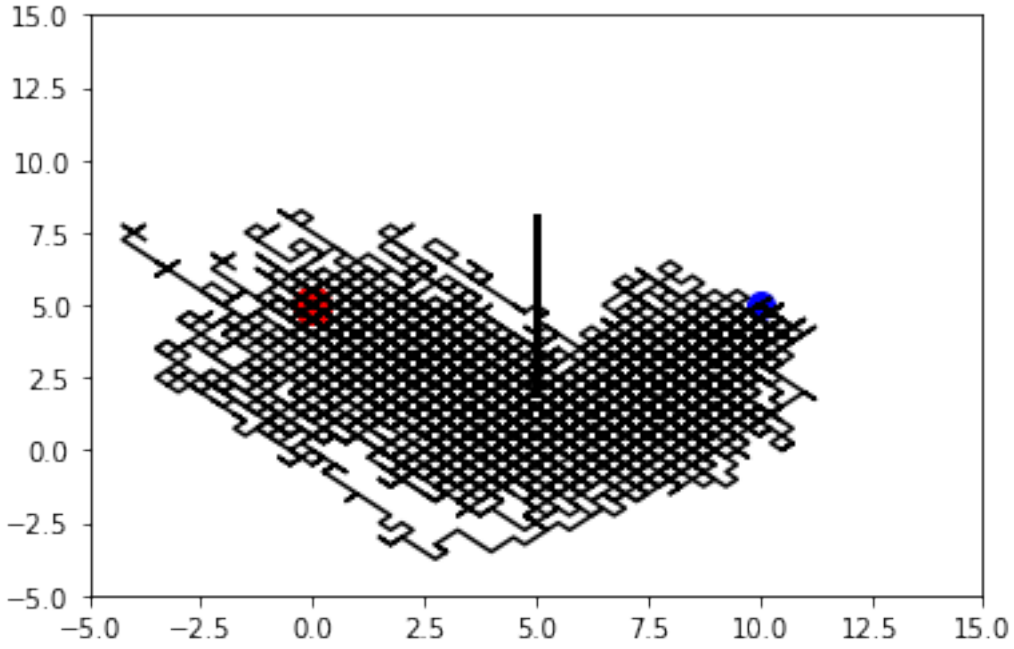

```
Ger: 18 | best fit: 0.545819714369
```



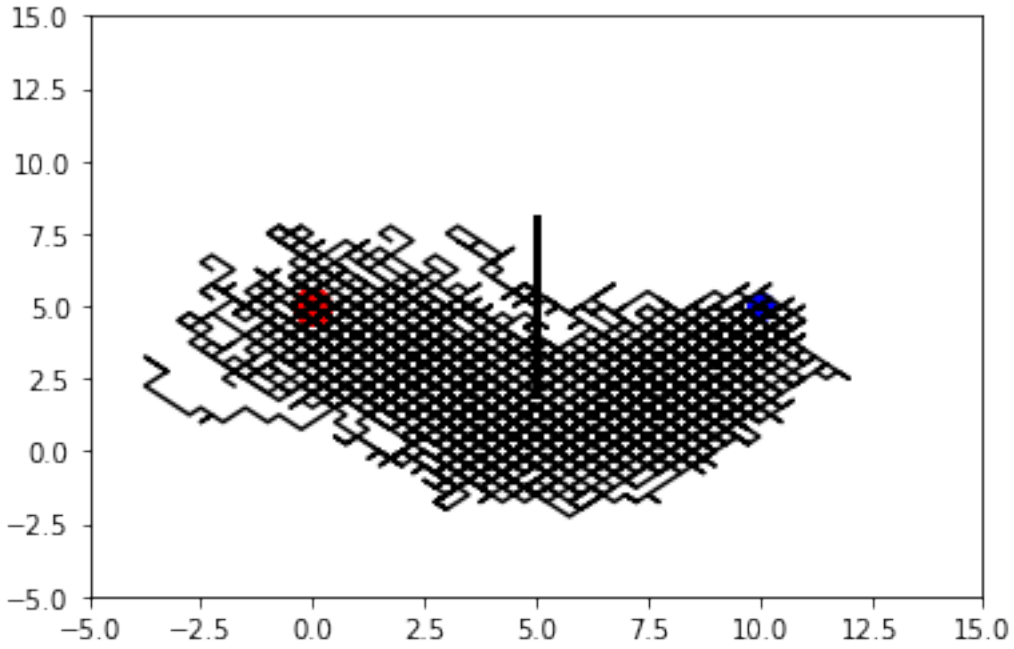
```
Ger: 19 | best fit: 0.545819714369
```



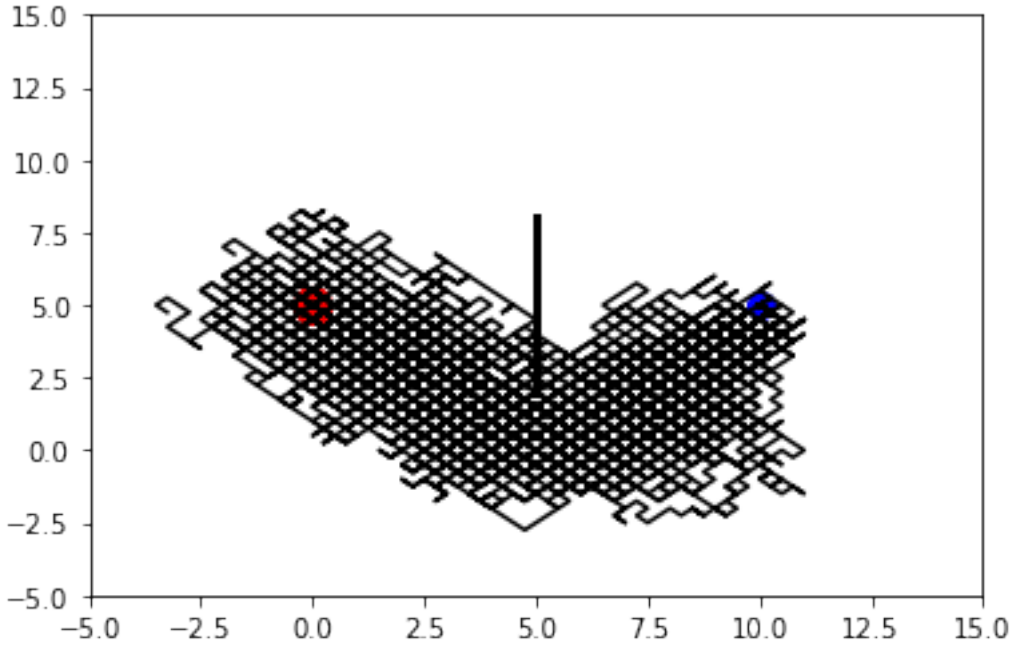

```
Ger: 20 | best fit: 0.545819714369
```



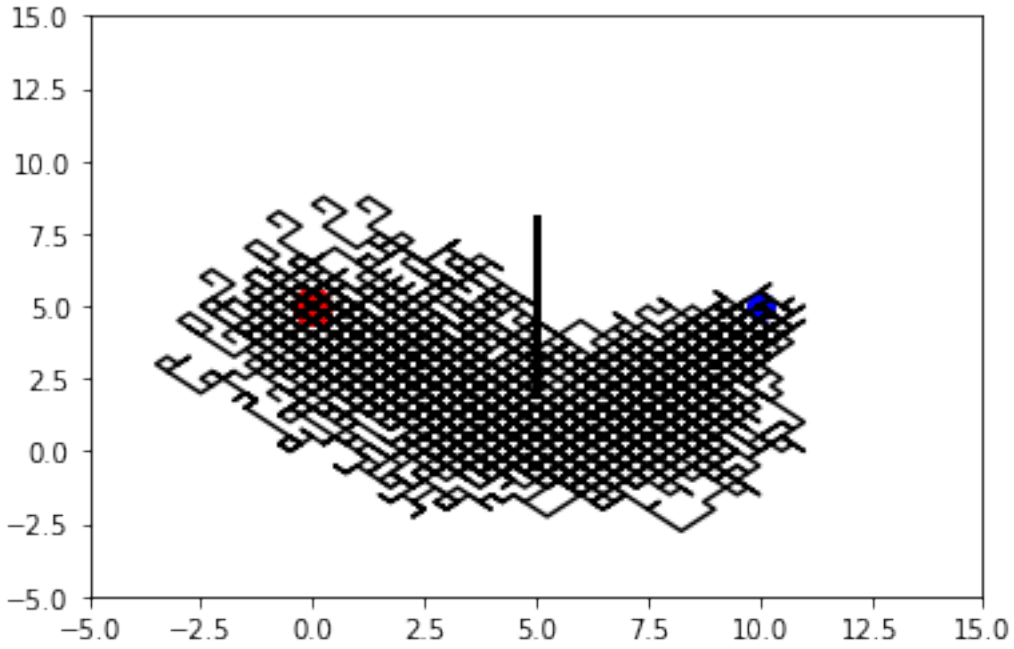
```
Ger: 21 | best fit: 0.545819714369
```



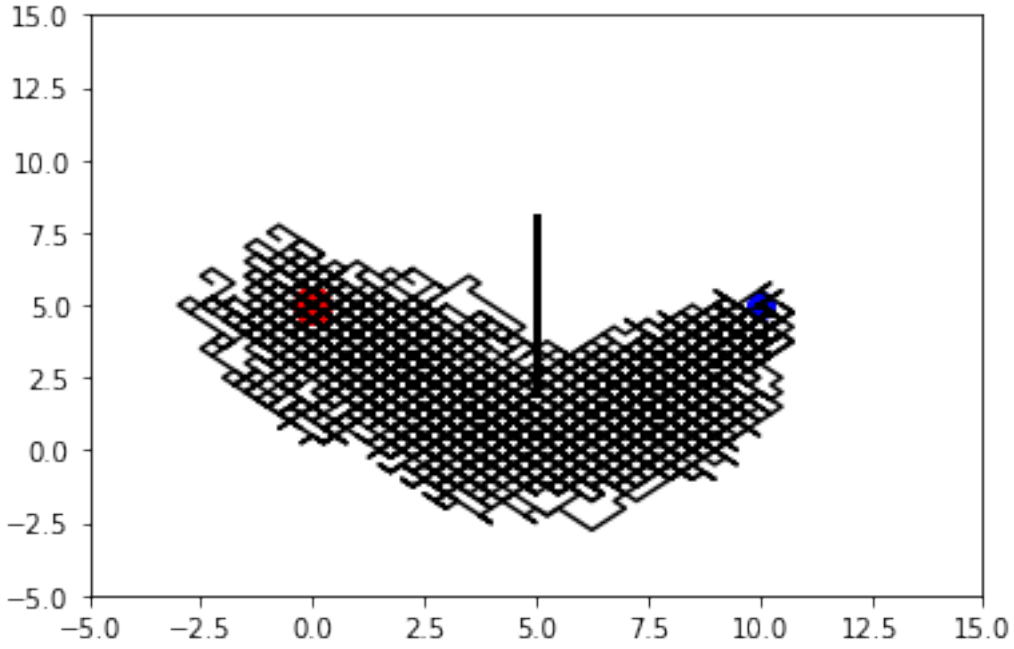

```
Ger: 22 | best fit: 0.545819714369
```



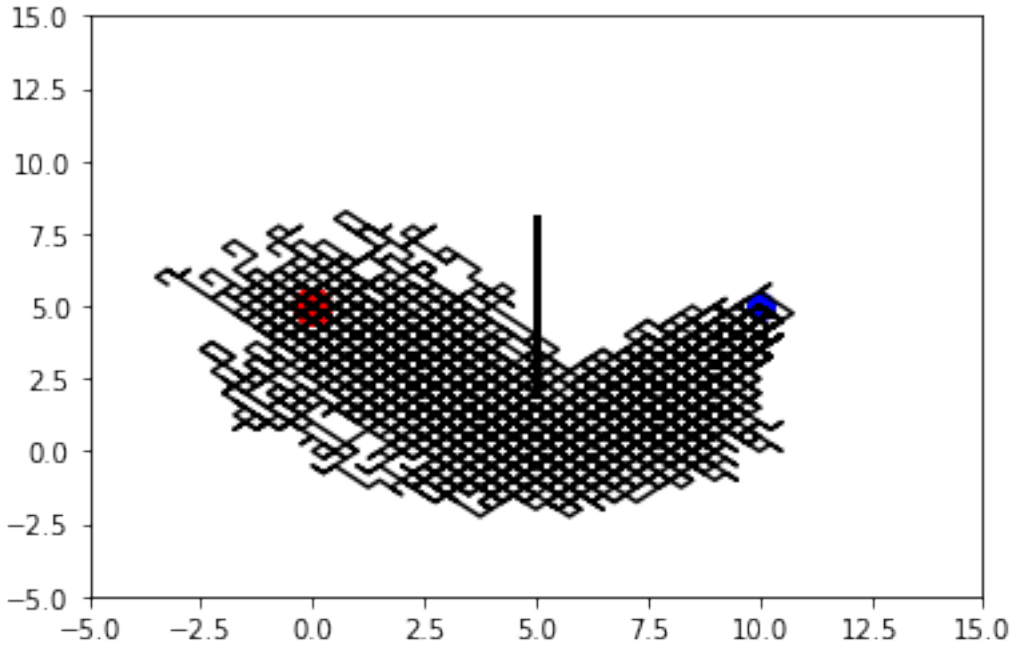
```
Ger: 23 | best fit: 0.545819714369
```



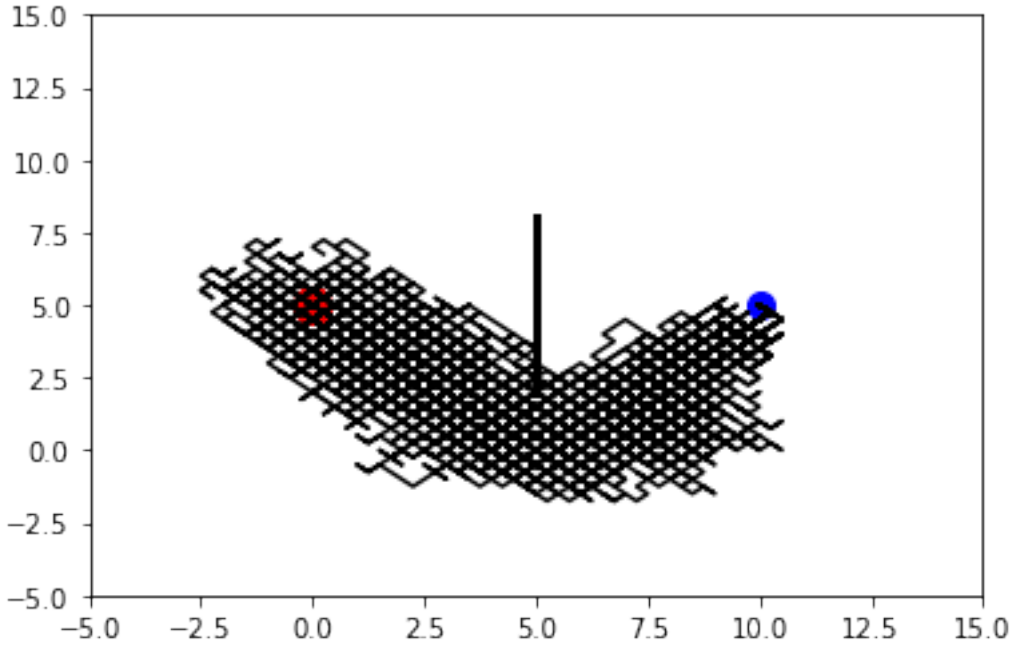

```
Ger: 24 | best fit: 0.545819714369
```



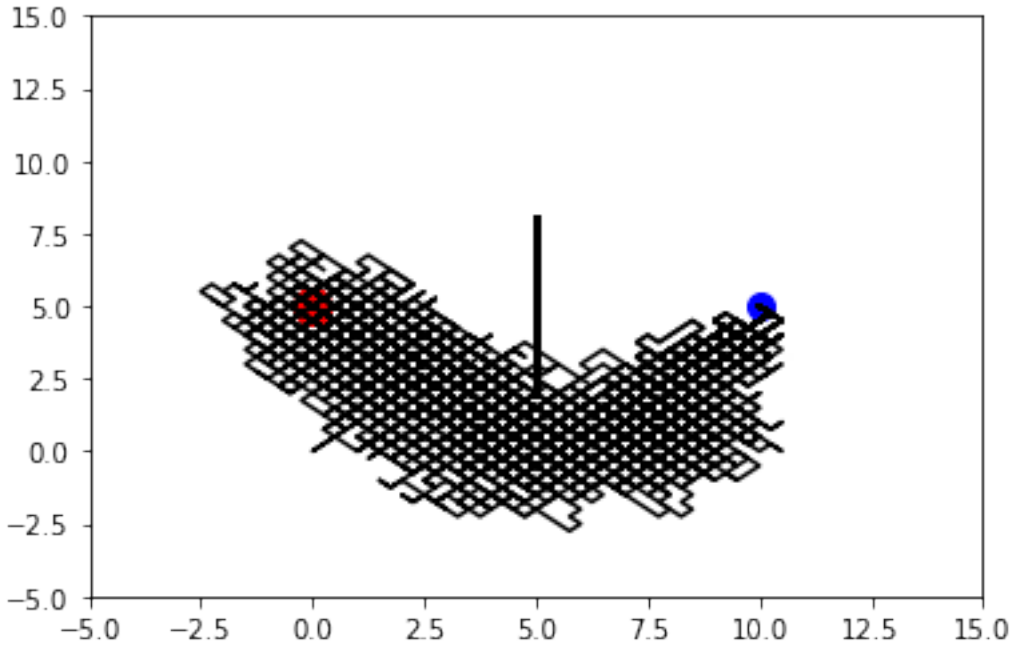
```
Ger: 25 | best fit: 0.545819714369
```



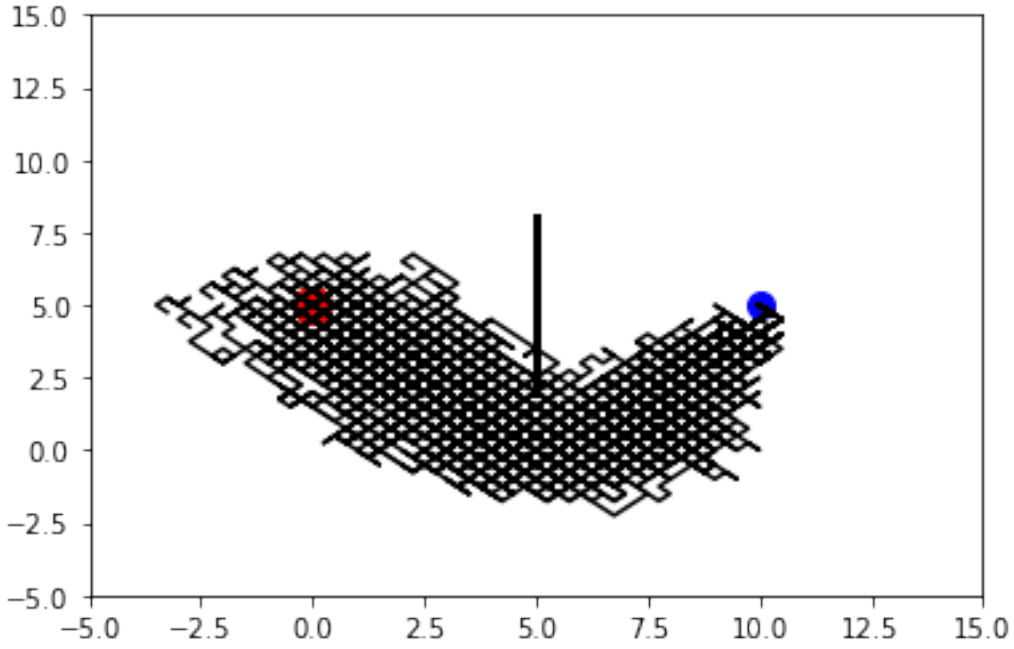

```
Ger: 26 | best fit: 0.545819714369
```



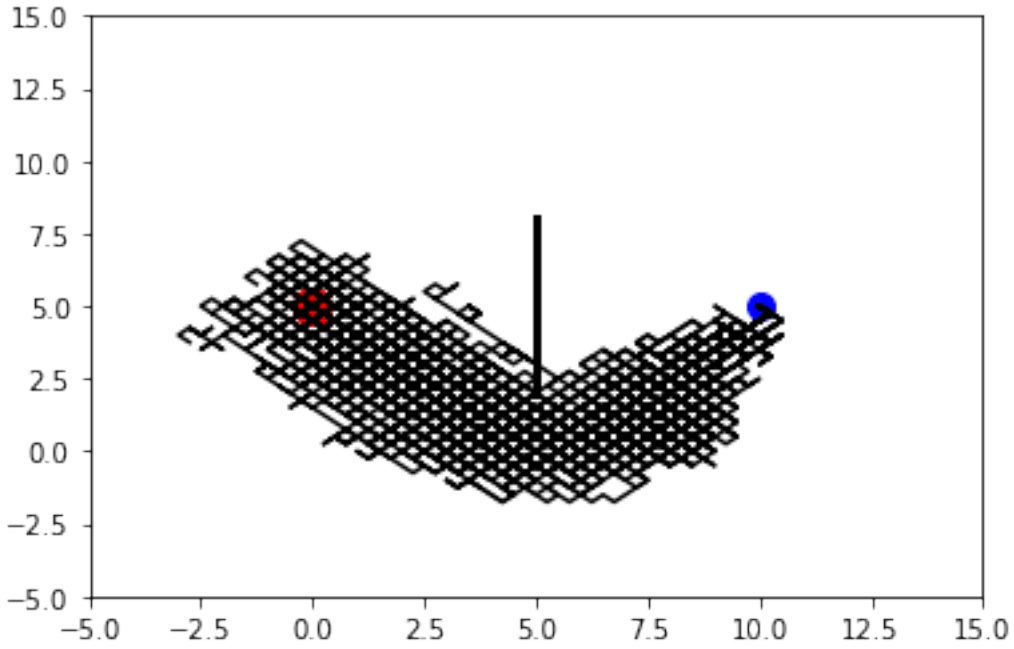
```
Ger: 27 | best fit: 0.545819714369
```



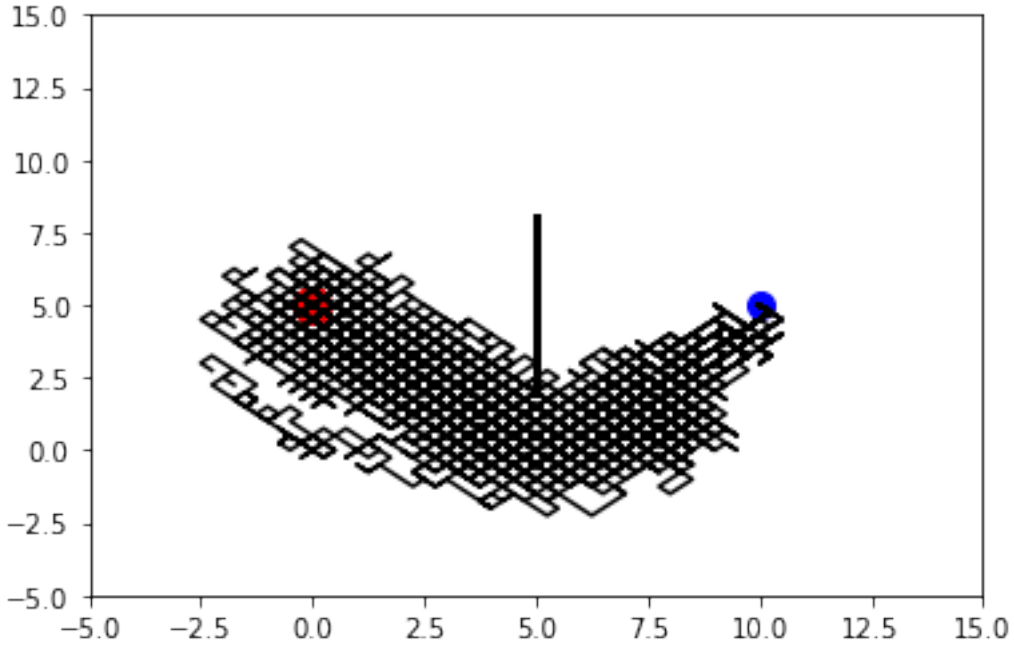

```
Ger: 28 | best fit: 0.545819714369
```



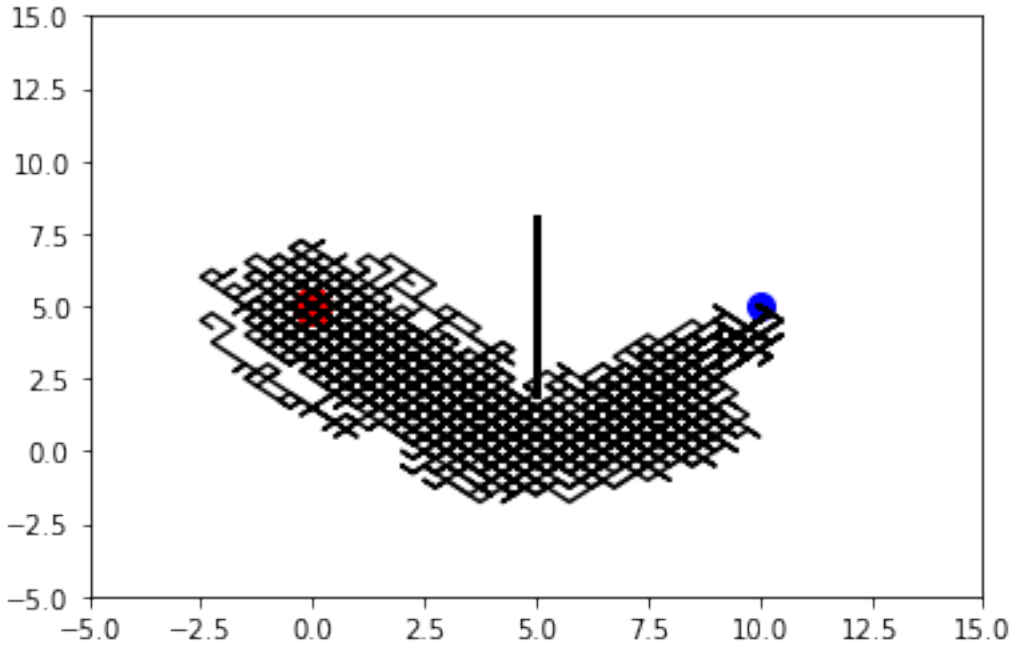
```
Ger: 29 | best fit: 0.545819714369
```



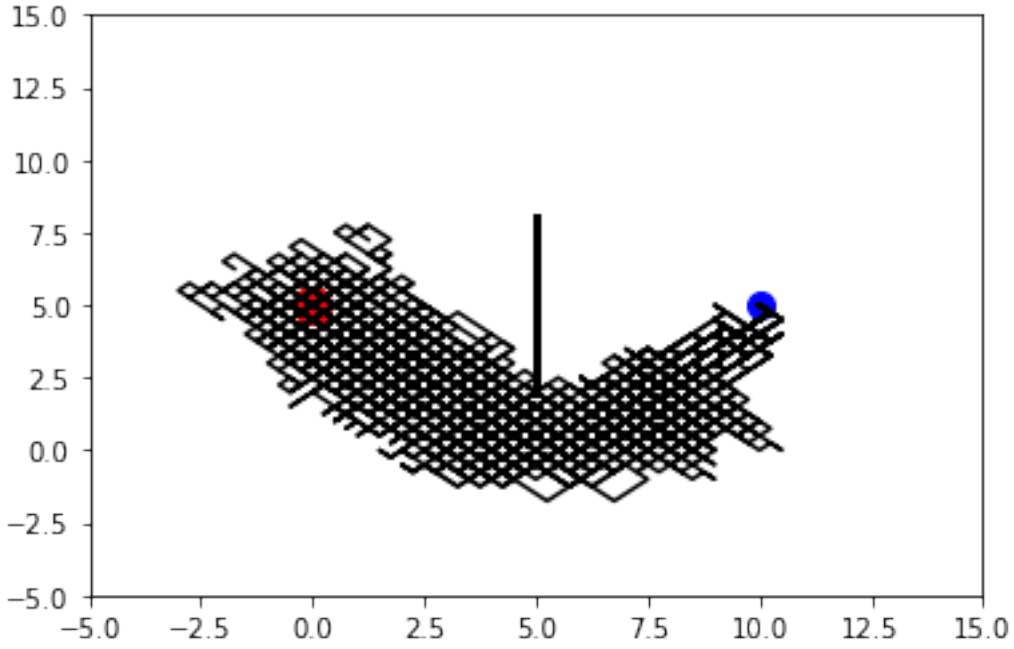

```
Ger: 30 | best fit: 0.545819714369
```



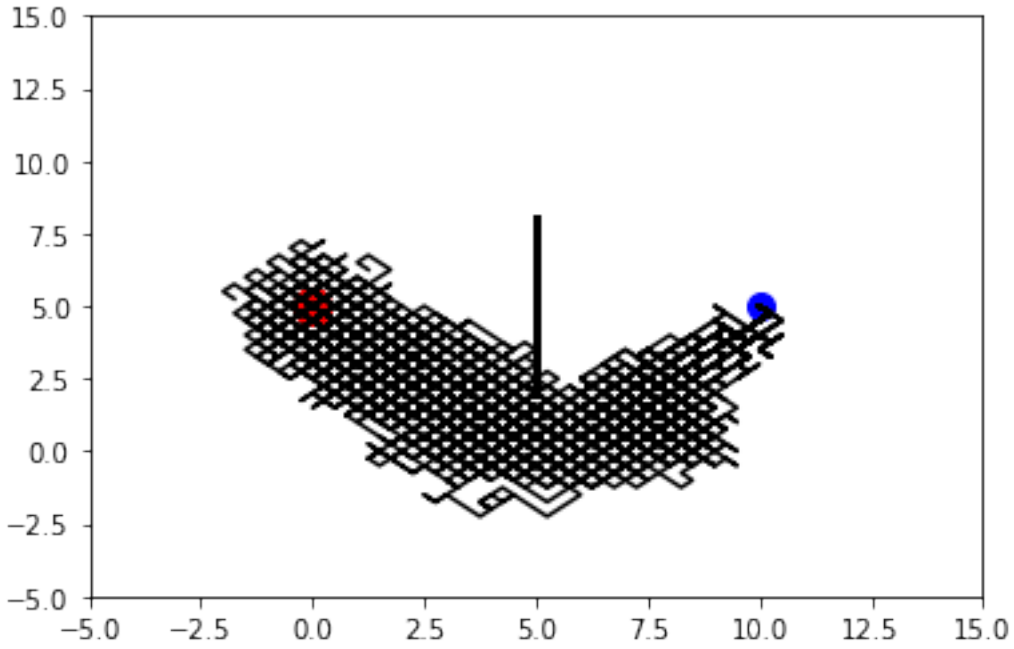
```
Ger: 31 | best fit: 0.545819714369
```



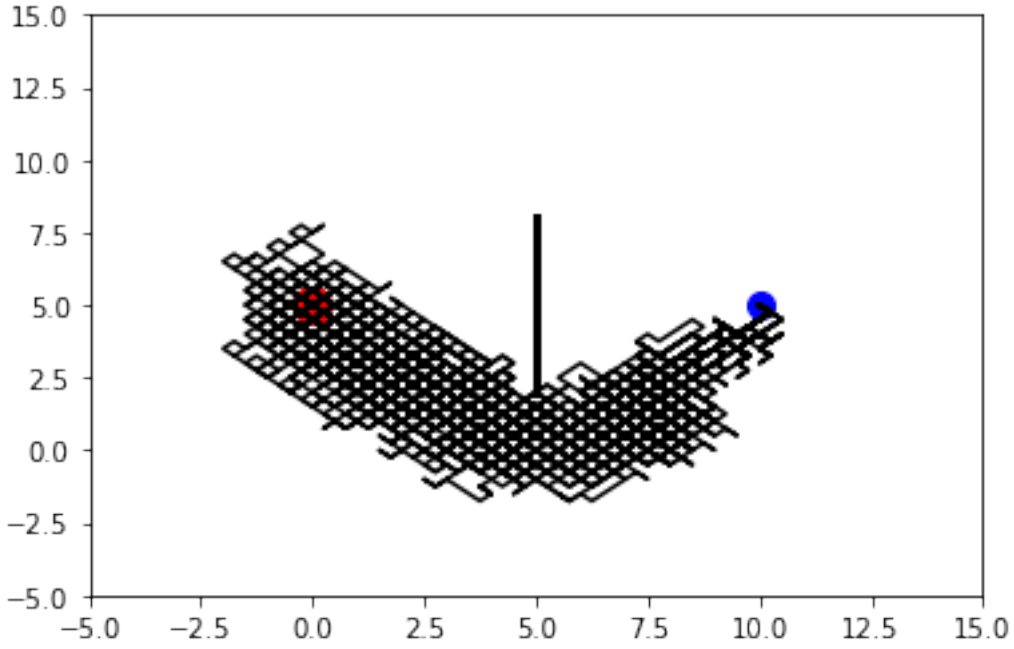

```
Ger: 32 | best fit: 0.545819714369
```



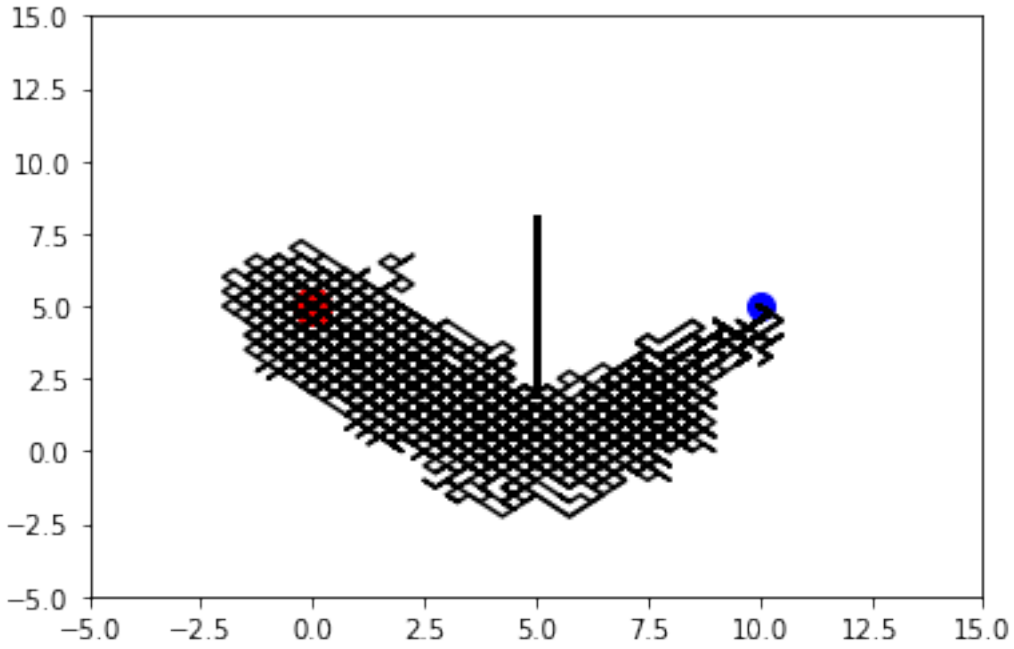
```
Ger: 33 | best fit: 0.545819714369
```



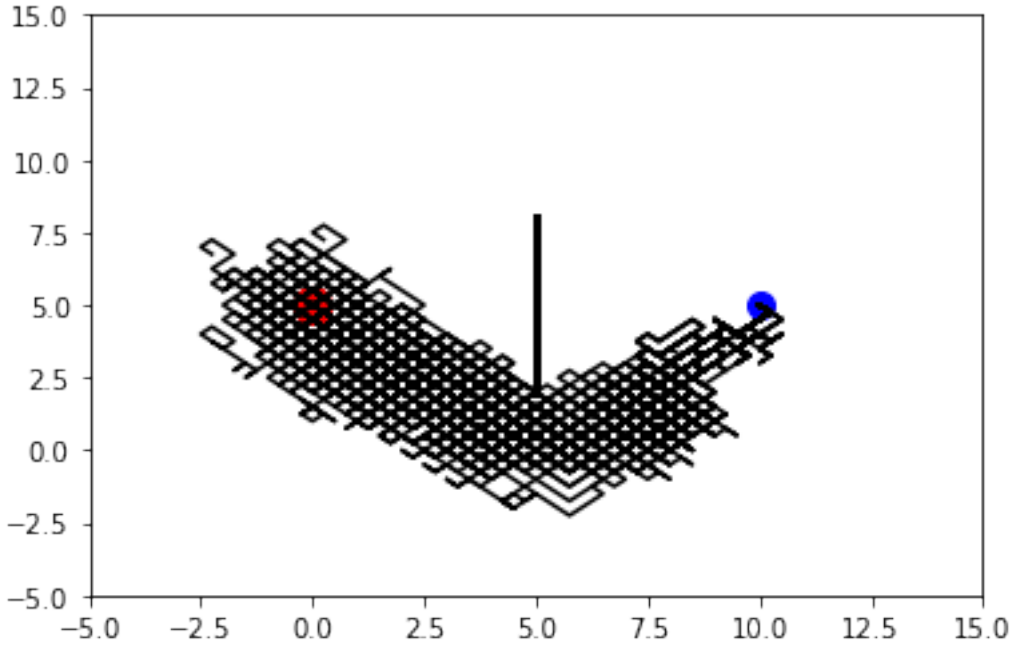

```
Ger: 34 | best fit: 0.545819714369
```



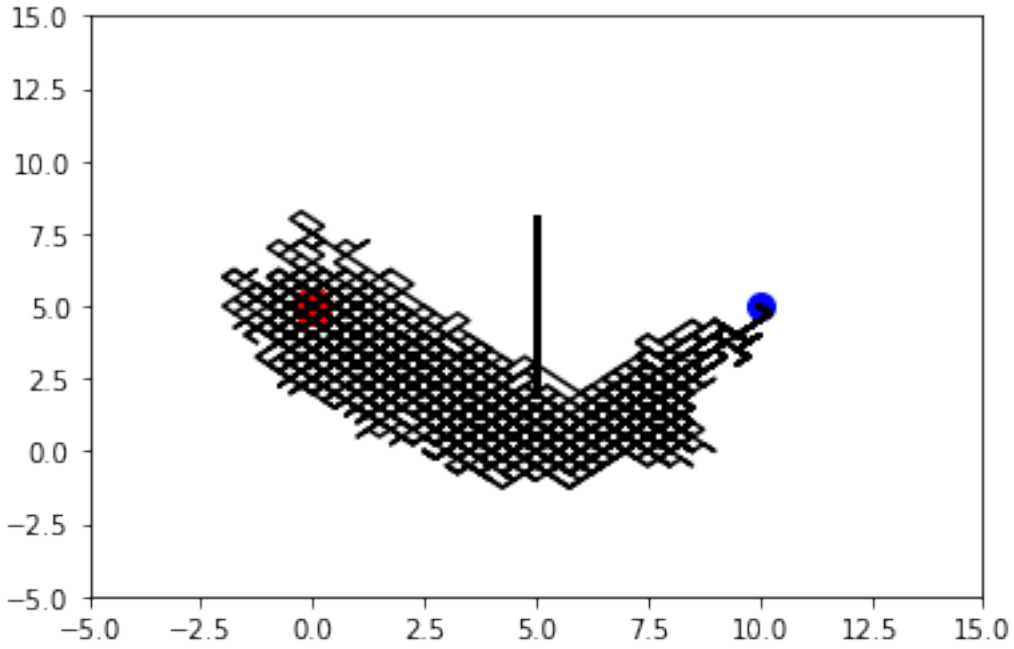
```
Ger: 35 | best fit: 0.545819714369
```



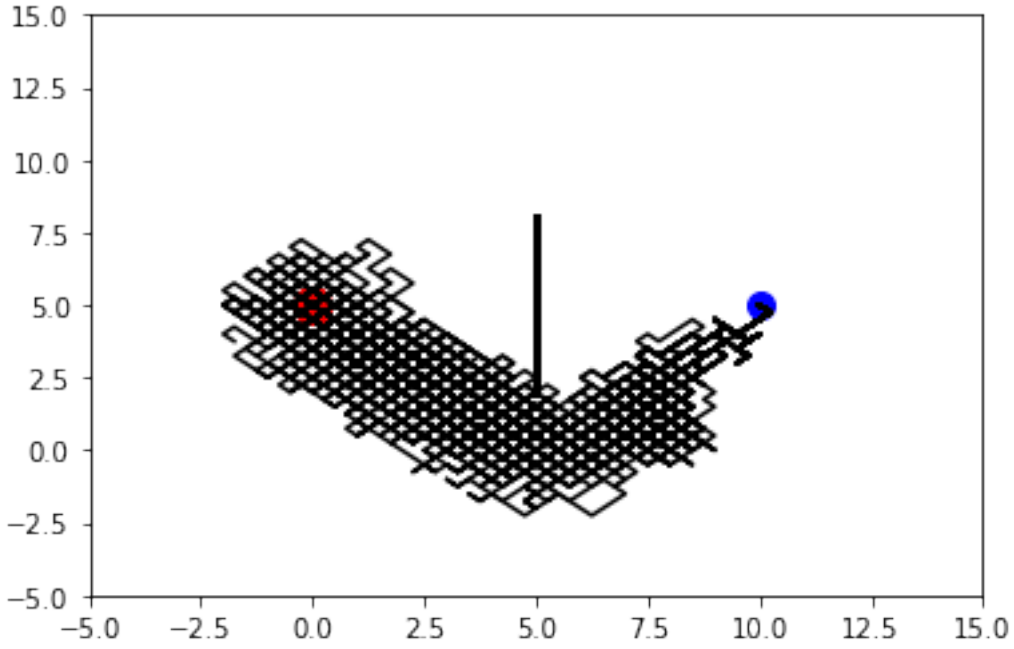

```
Ger: 36 | best fit: 0.545819714369
```



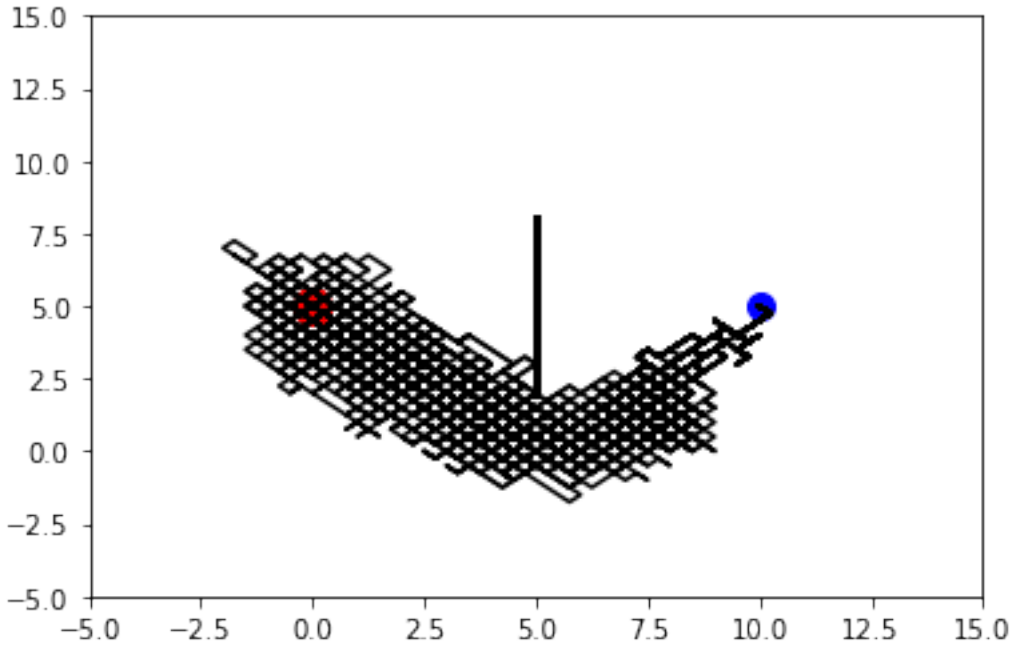
```
Ger: 37 | best fit: 0.545819714369
```



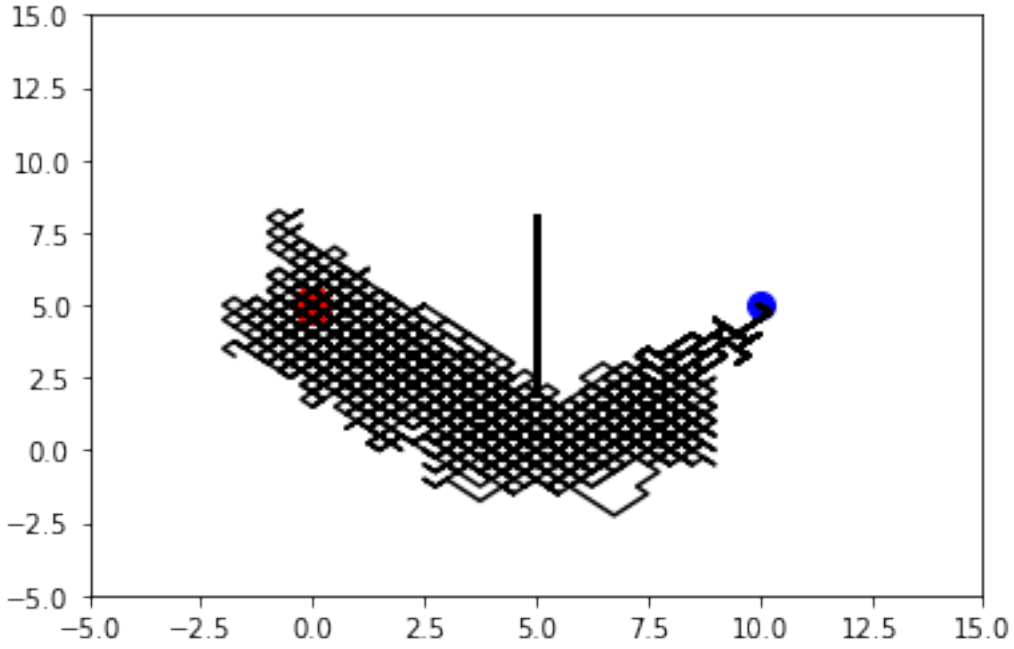

```
Ger: 38 | best fit: 0.545819714369
```



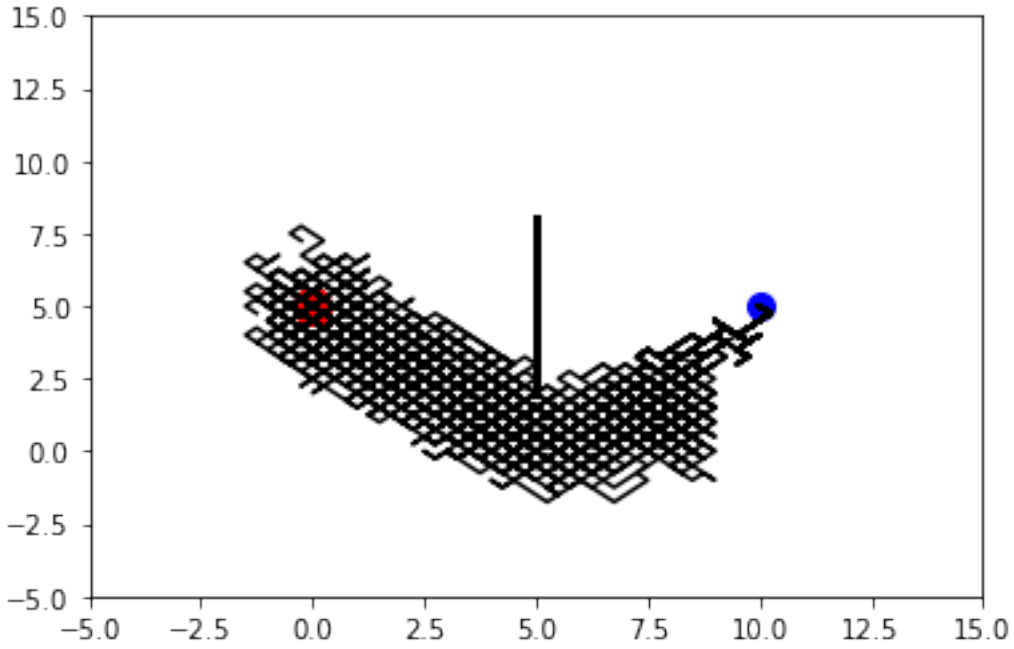
```
Ger: 39 | best fit: 0.545819714369
```



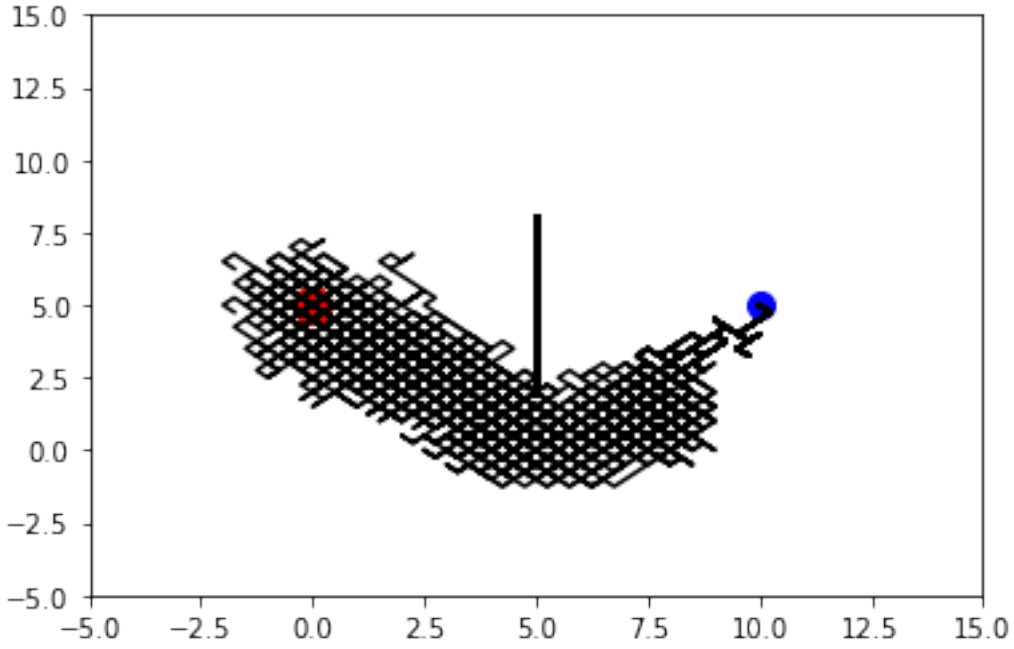

```
Ger: 40 | best fit: 0.545819714369
```



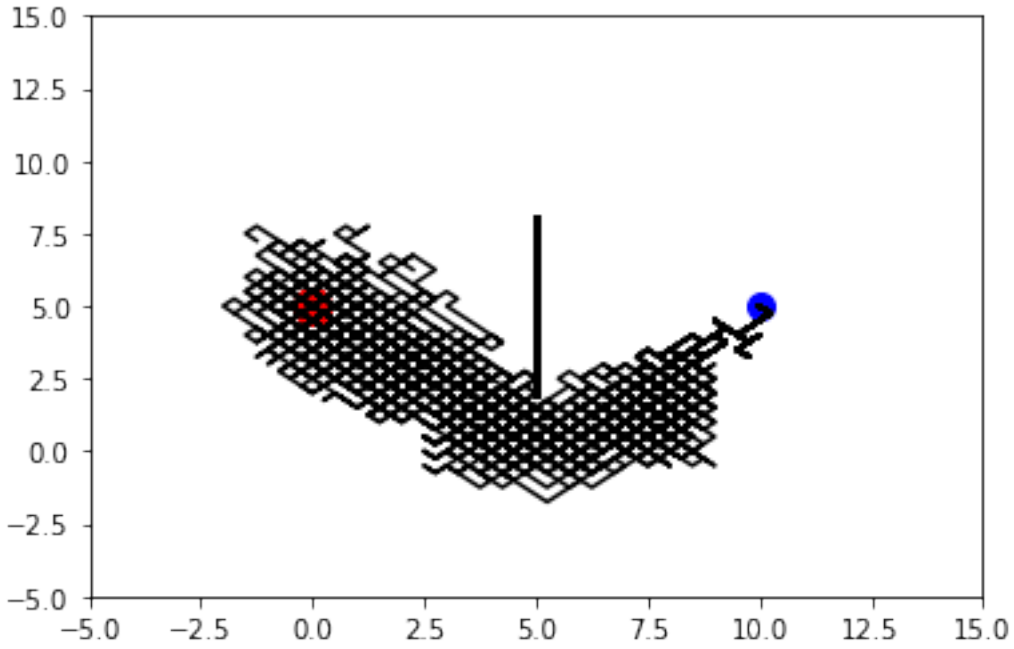
```
Ger: 41 | best fit: 0.545819714369
```



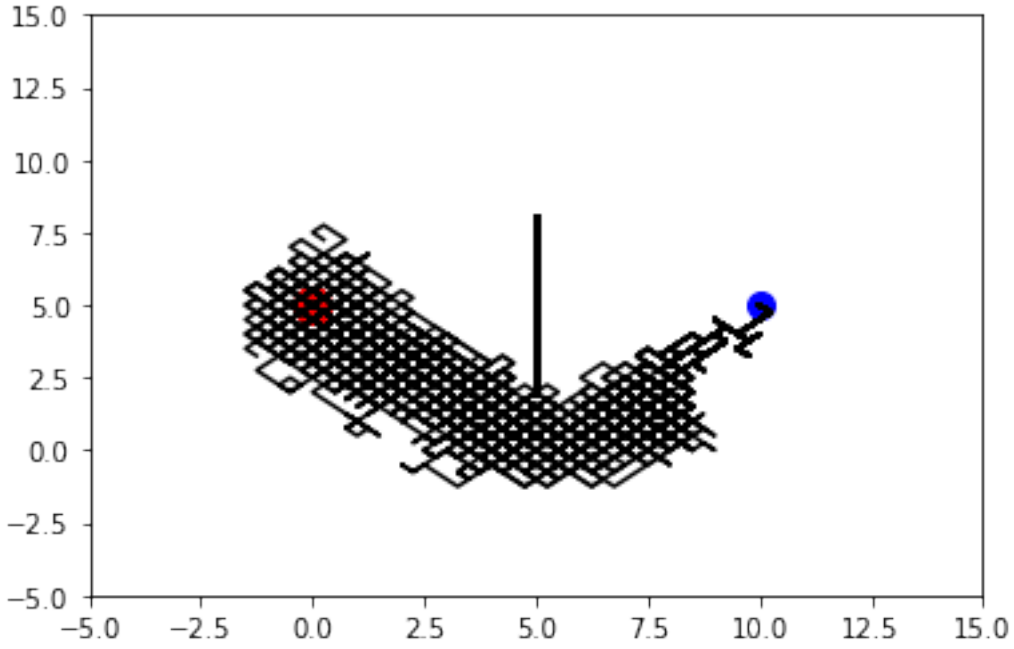

```
Ger: 42 | best fit: 0.545819714369
```



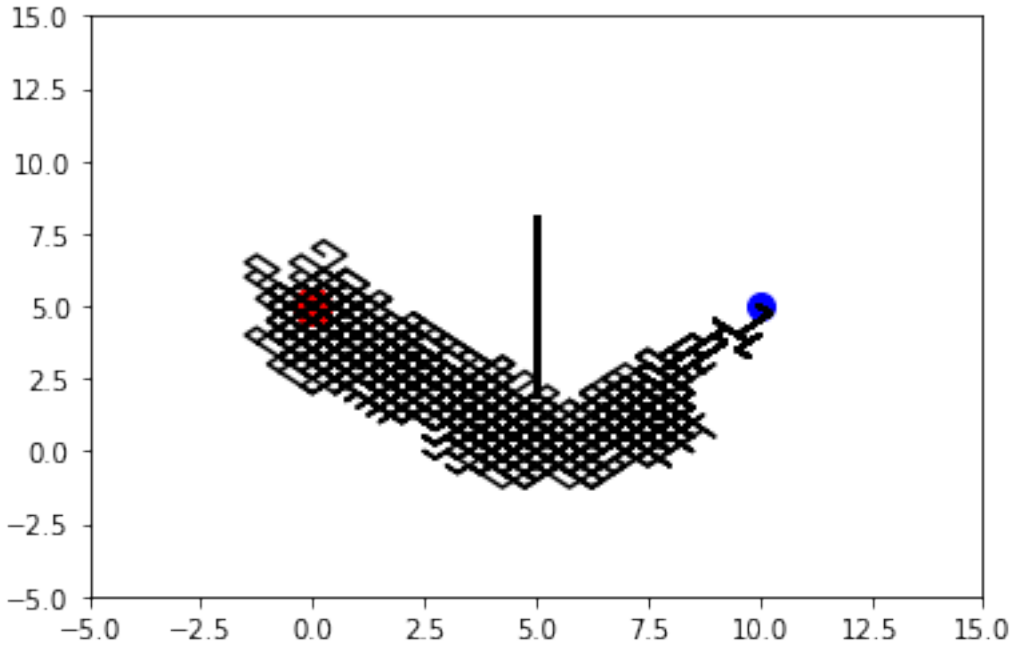
```
Ger: 43 | best fit: 0.545819714369
```



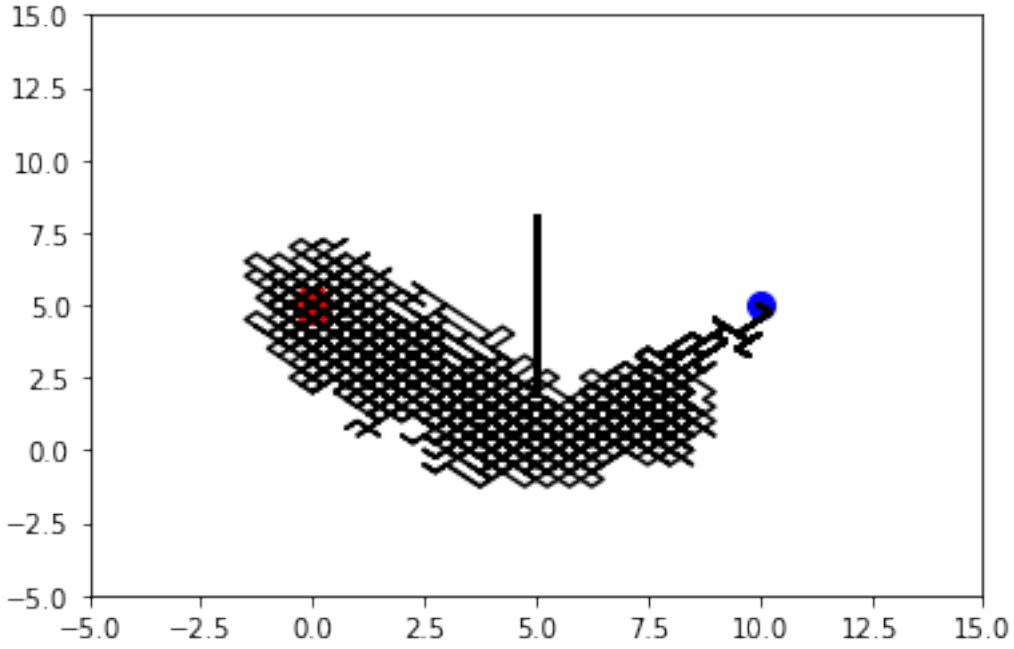

```
Ger: 44 | best fit: 0.545819714369
```



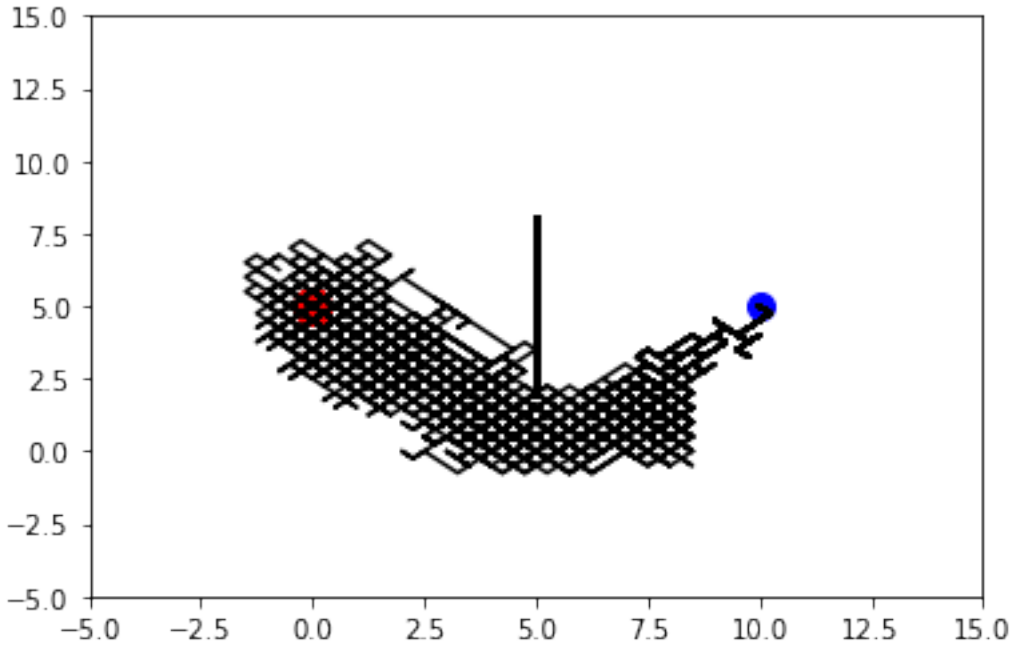
```
Ger: 45 | best fit: 0.545819714369
```



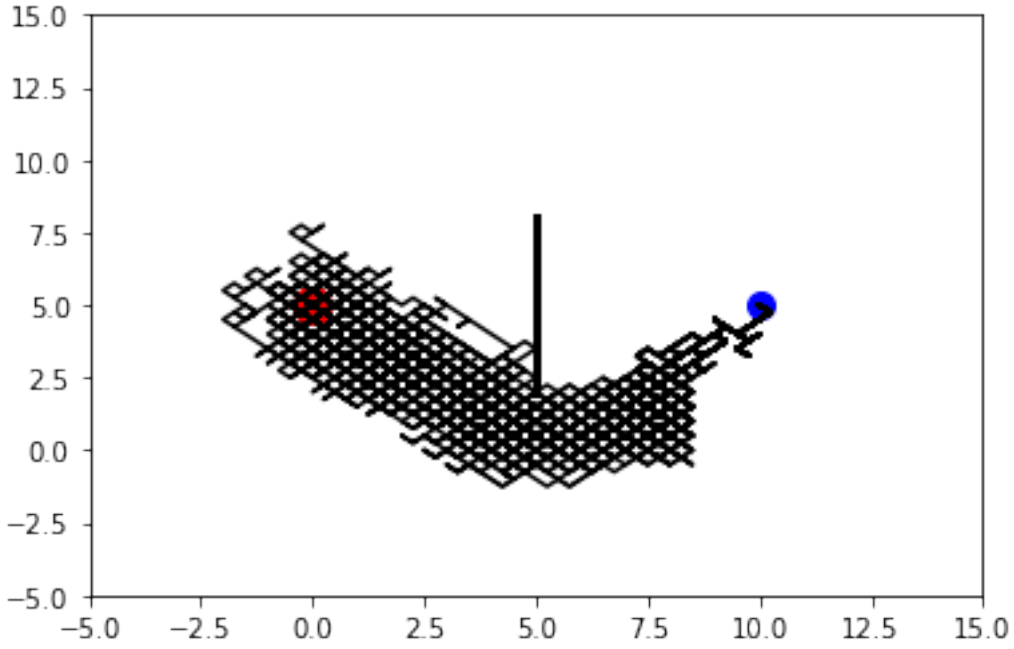

```
Ger: 46 | best fit: 0.545819714369
```



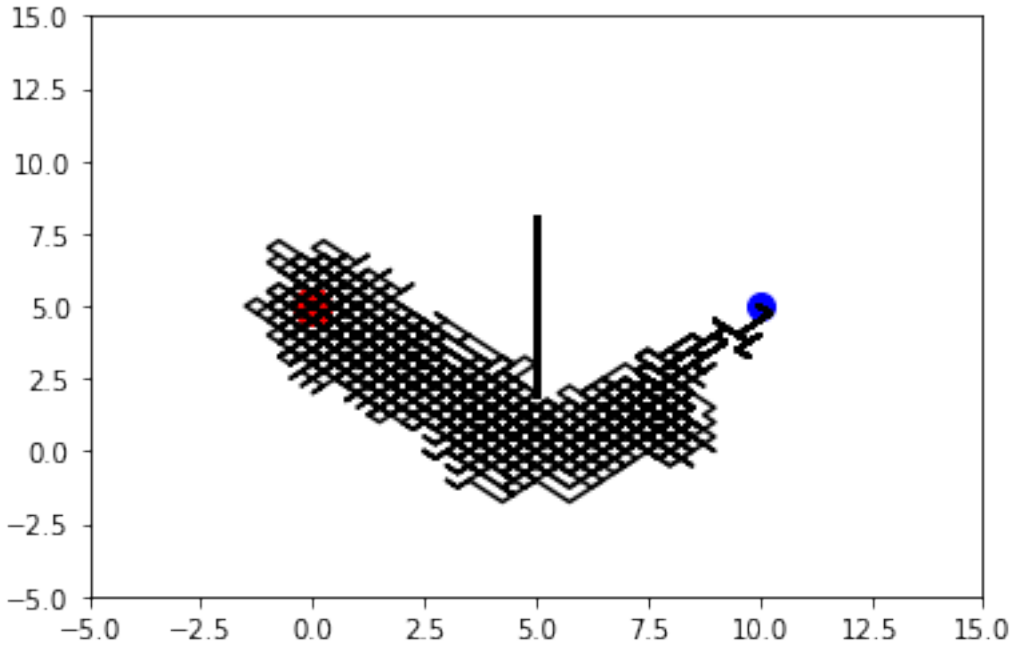
```
Ger: 47 | best fit: 0.545819714369
```



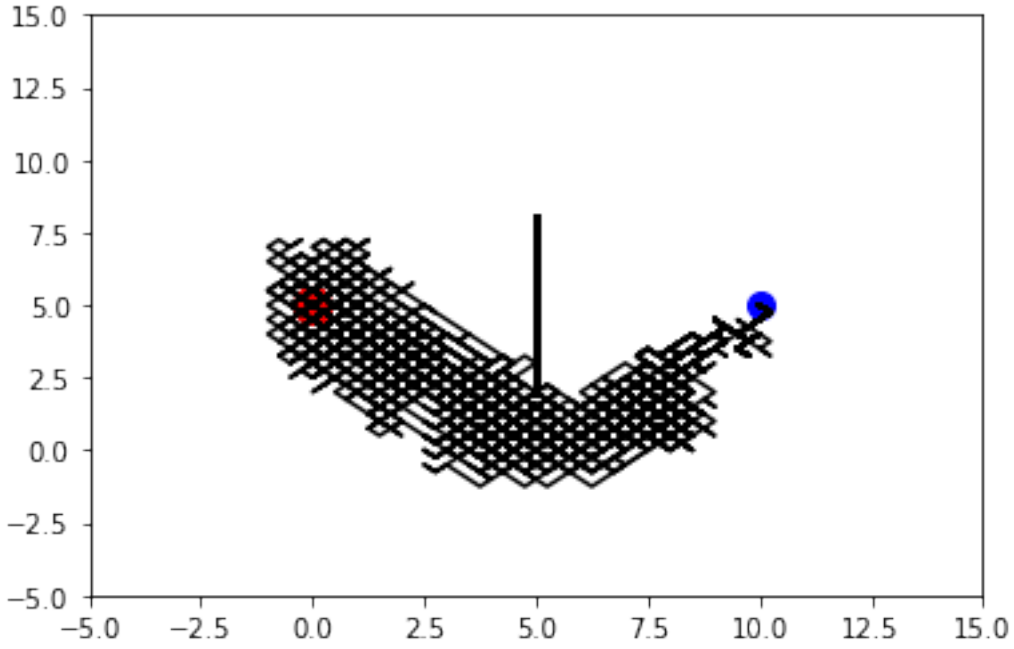

```
Ger: 48 | best fit: 0.545819714369
```



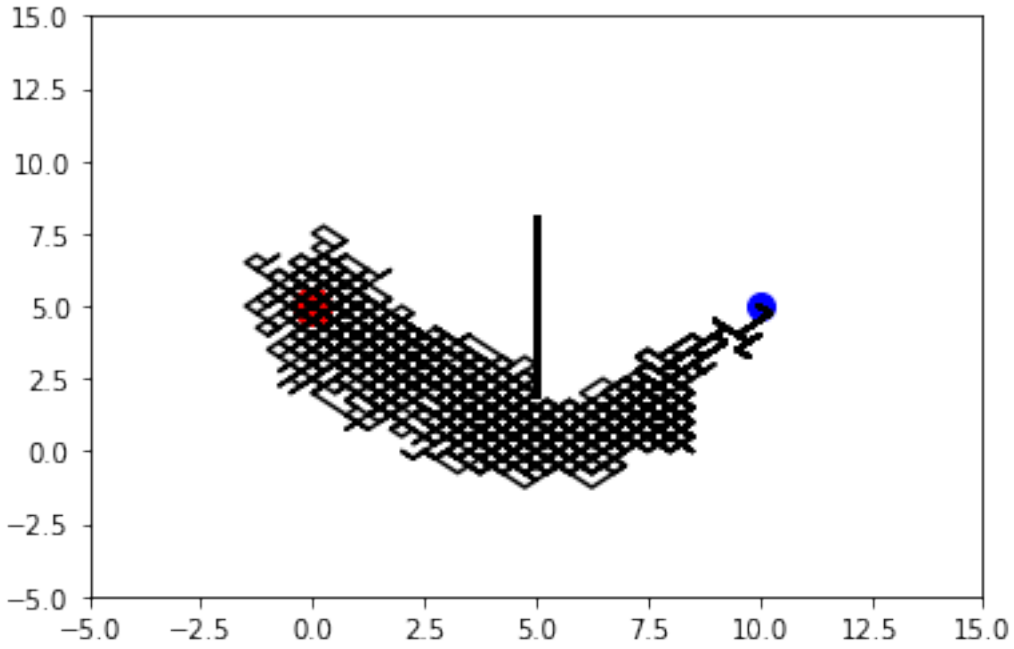
```
Ger: 49 | best fit: 0.545819714369
```



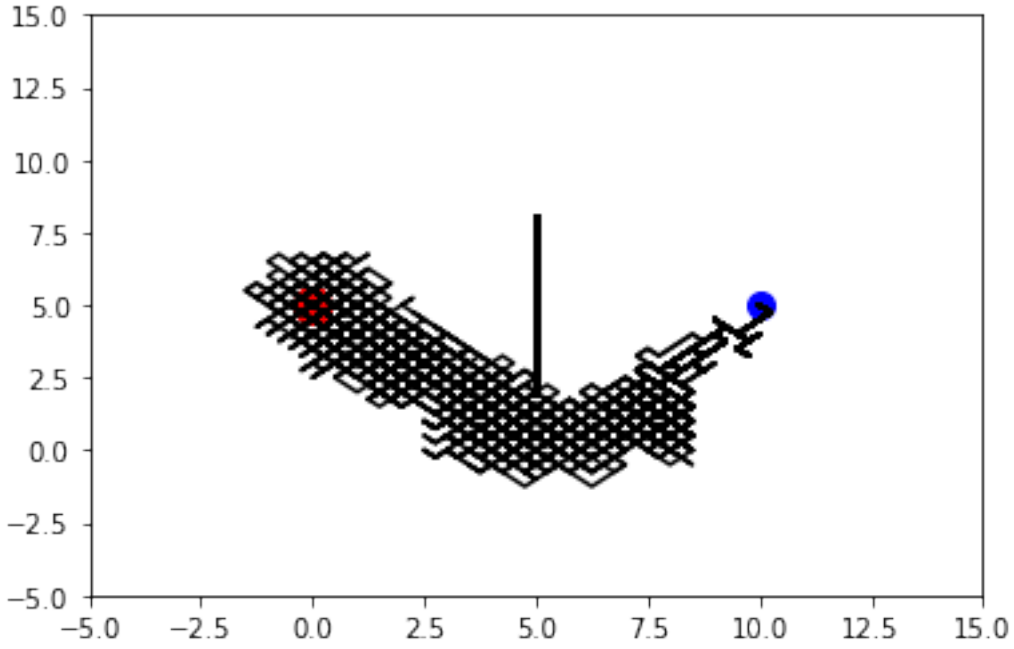

```
Ger: 50 | best fit: 0.545819714369
```



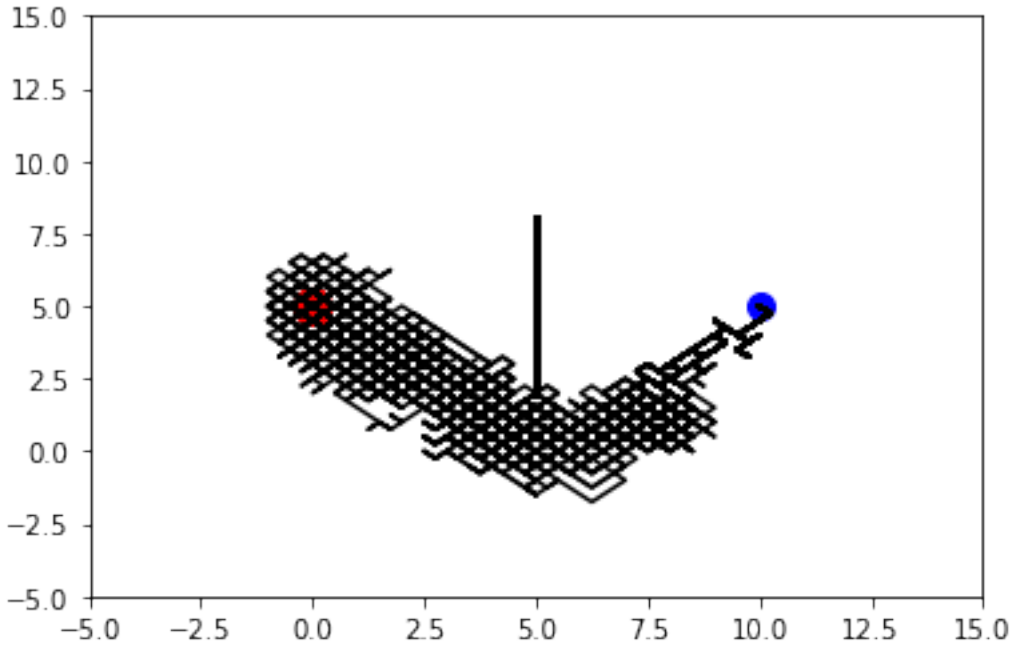
```
Ger: 51 | best fit: 0.545819714369
```



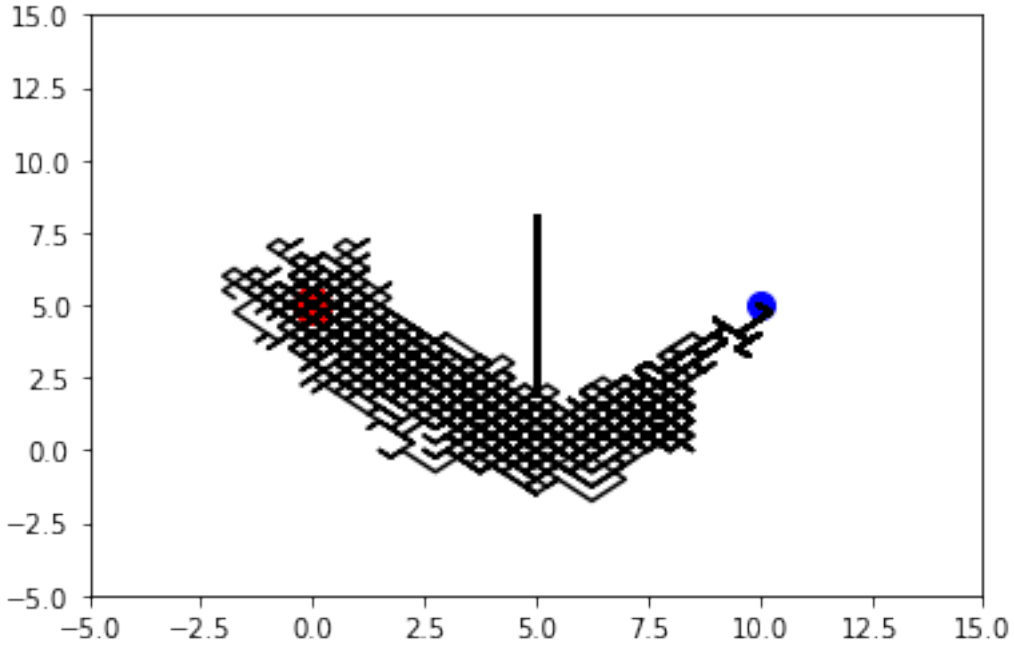

```
Ger: 52 | best fit: 0.545819714369
```



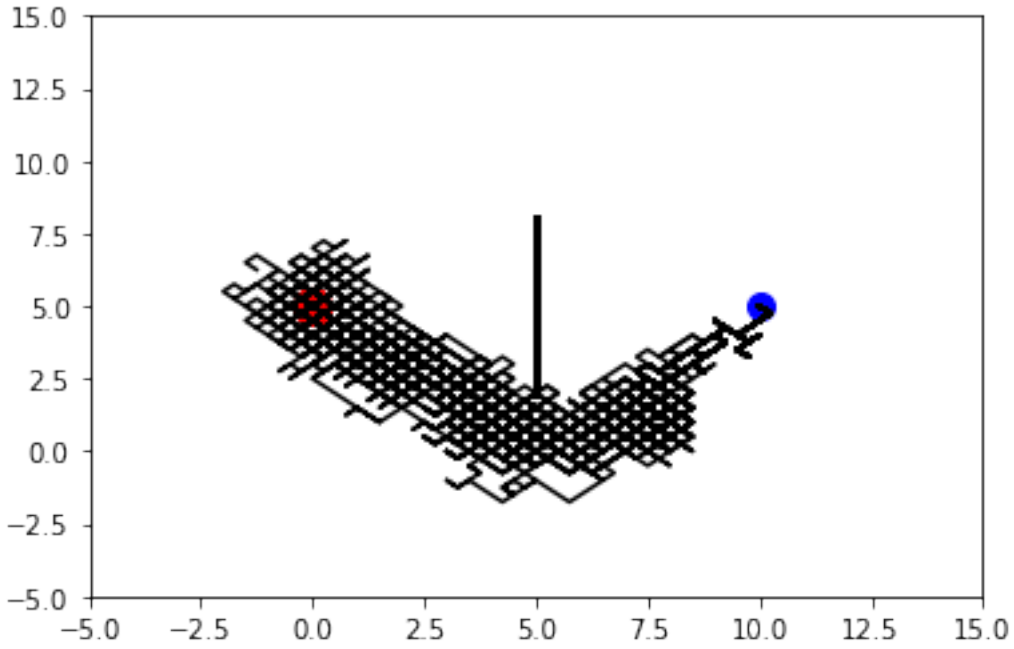
```
Ger: 53 | best fit: 0.545819714369
```



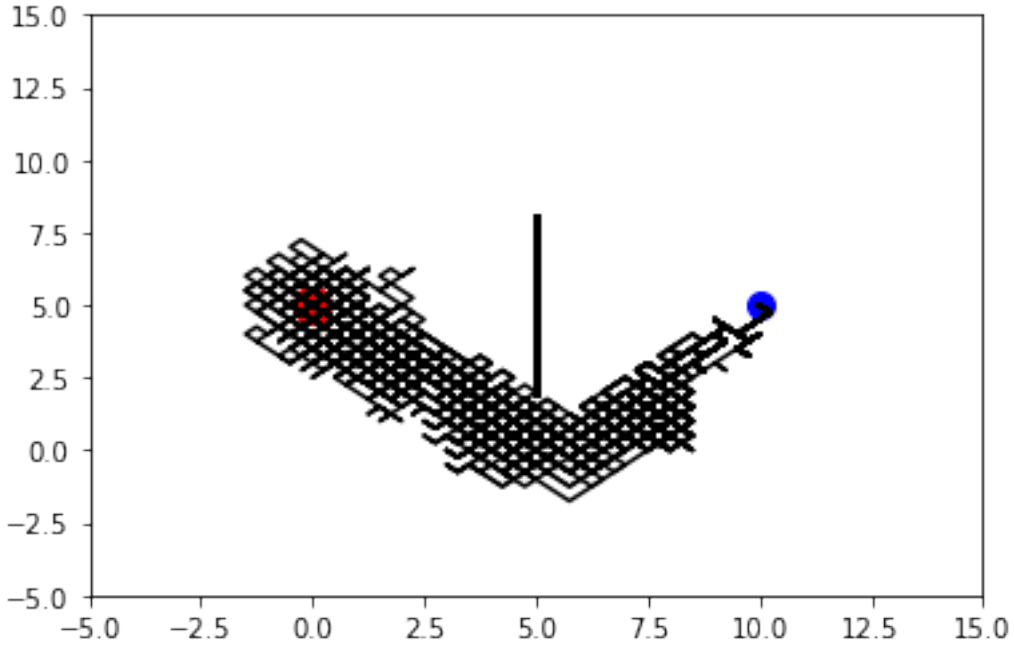

```
Ger: 54 | best fit: 0.545819714369
```



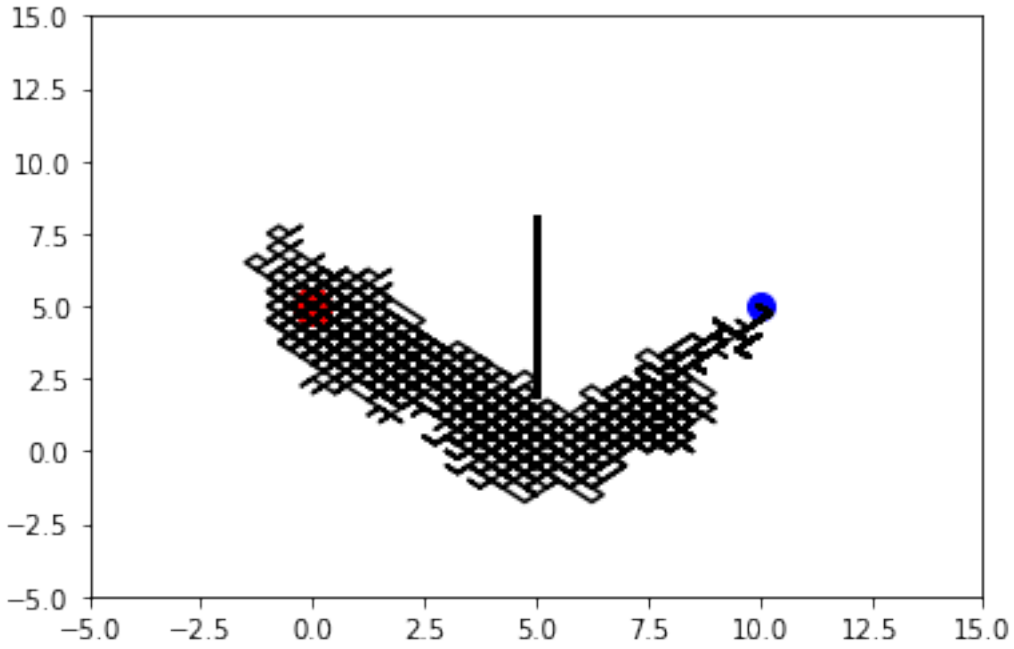
```
Ger: 55 | best fit: 0.545819714369
```



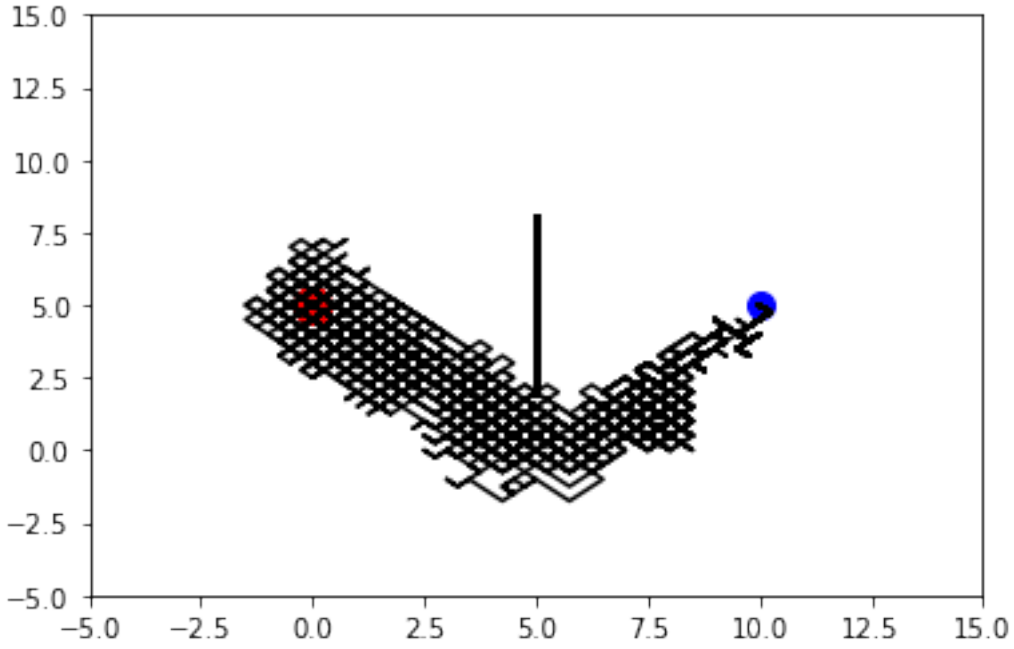

```
Ger: 56 | best fit: 0.545819714369
```



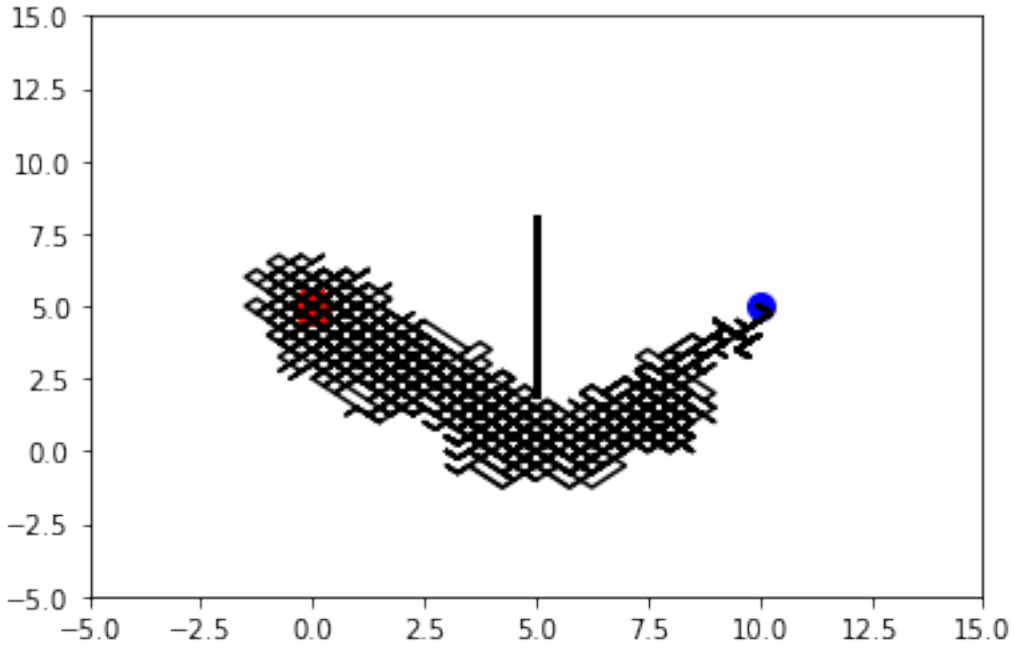
```
Ger: 57 | best fit: 0.545819714369
```



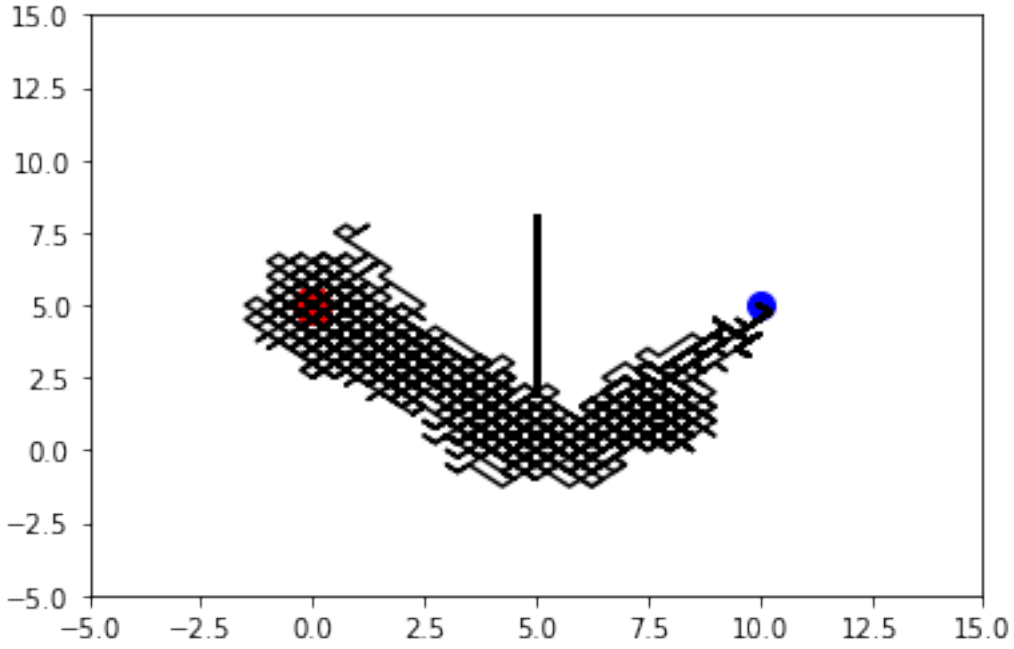

```
Ger: 58 | best fit: 0.545819714369
```



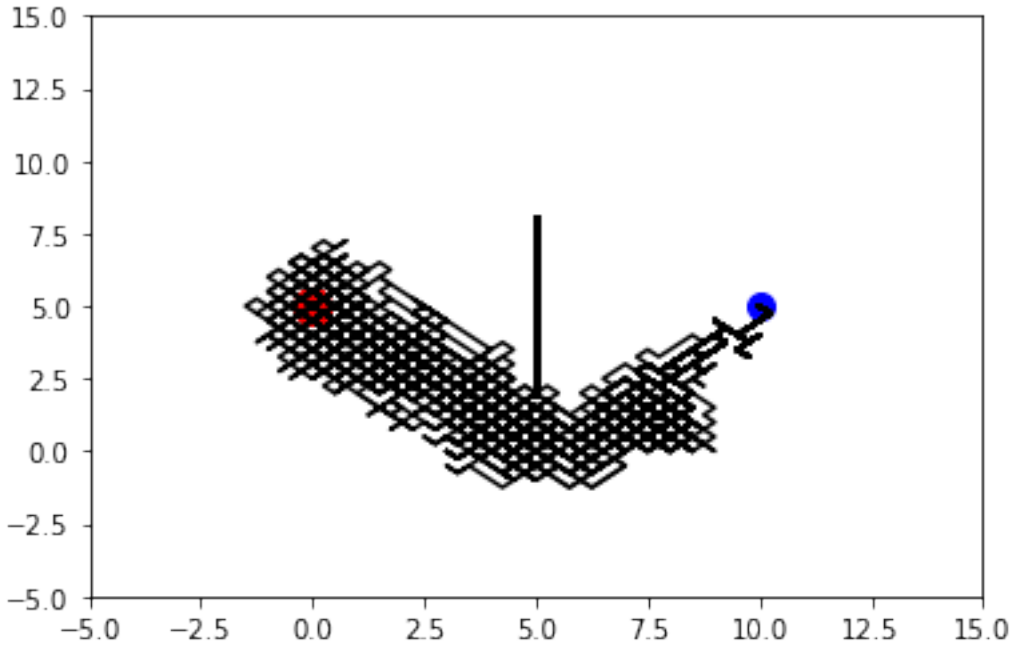
```
Ger: 59 | best fit: 0.545819714369
```



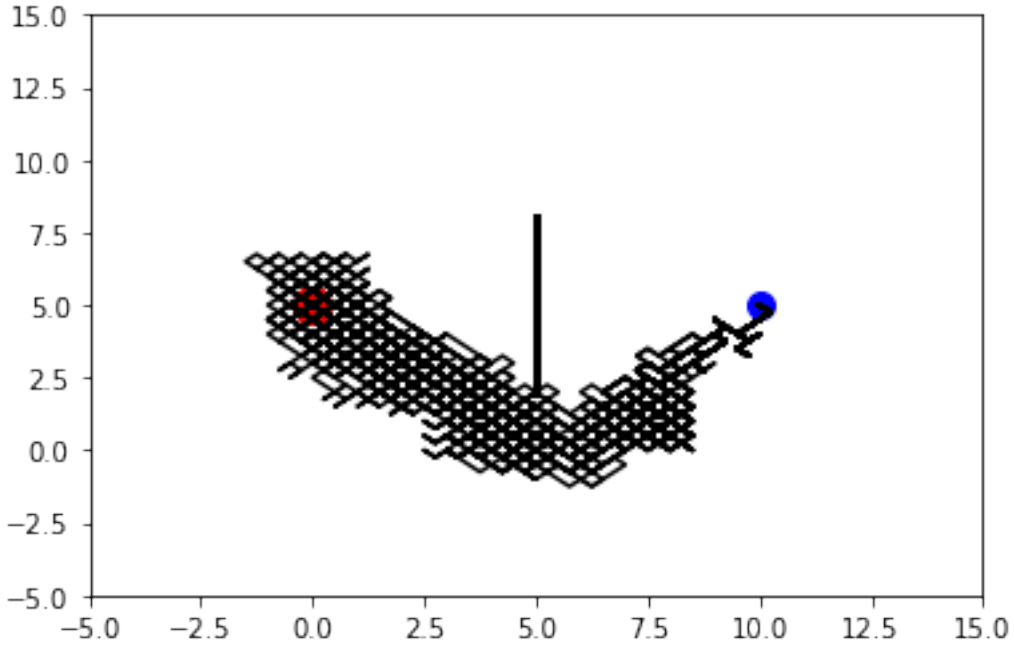

```
Ger: 60 | best fit: 0.545819714369
```



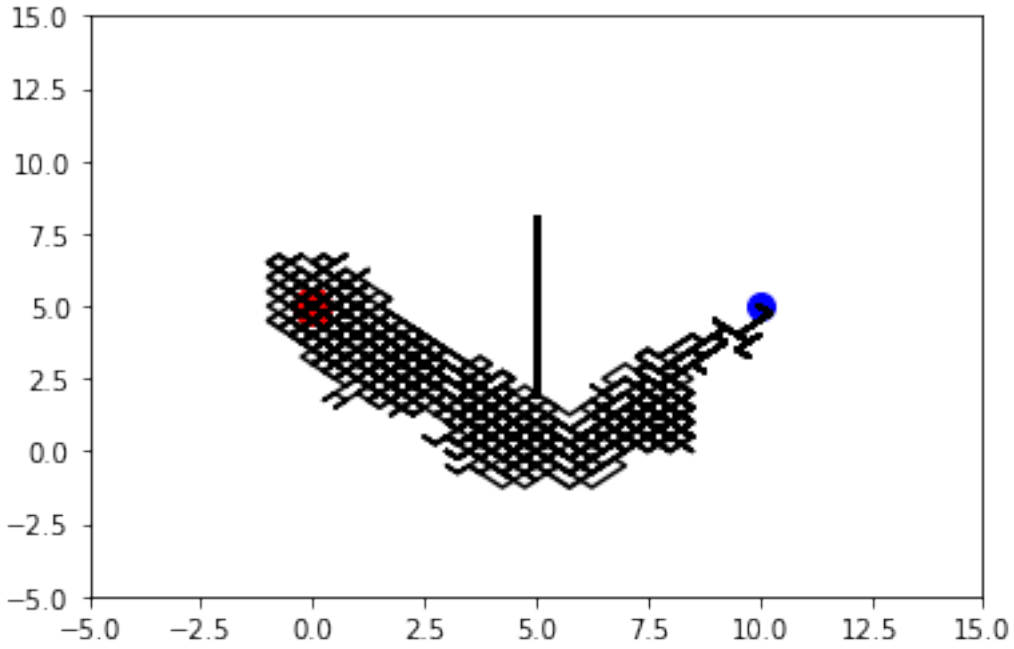
```
Ger: 61 | best fit: 0.545819714369
```



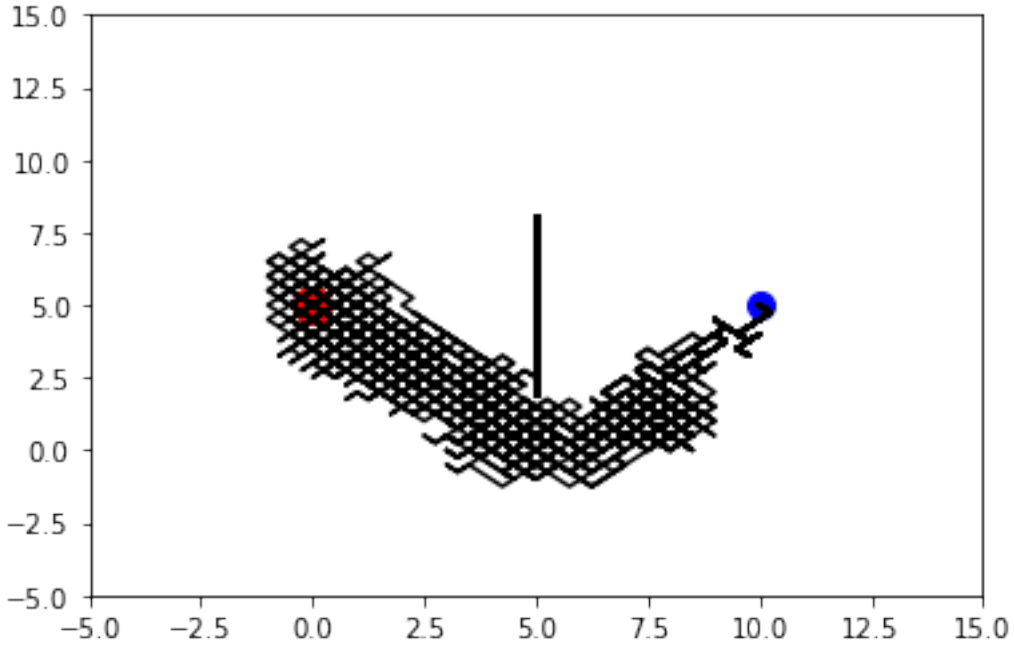

```
Ger: 62 | best fit: 0.545819714369
```



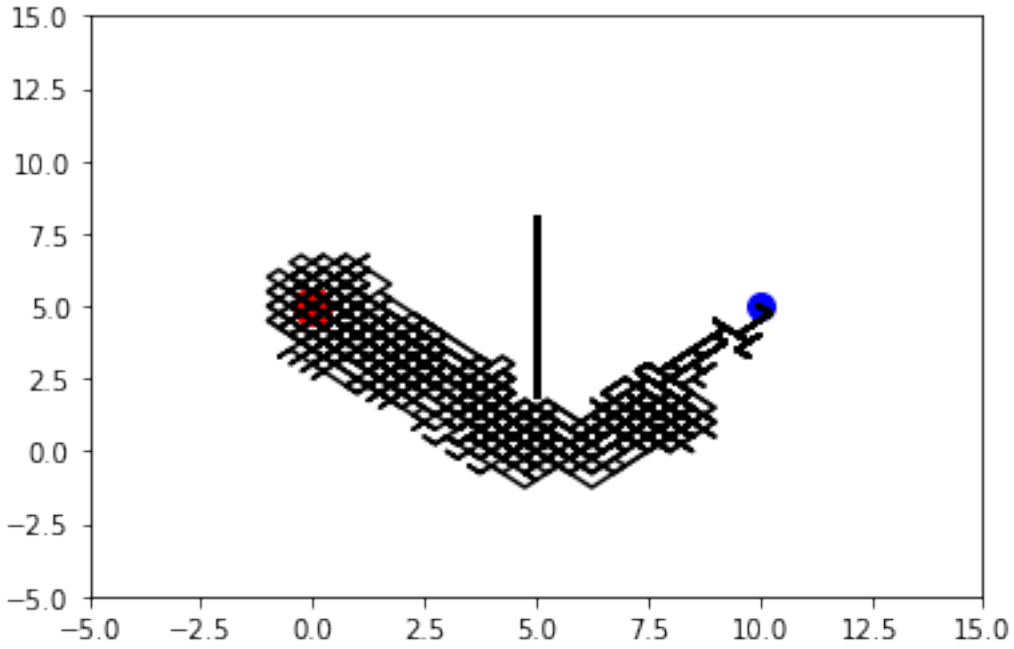
```
Ger: 63 | best fit: 0.545819714369
```



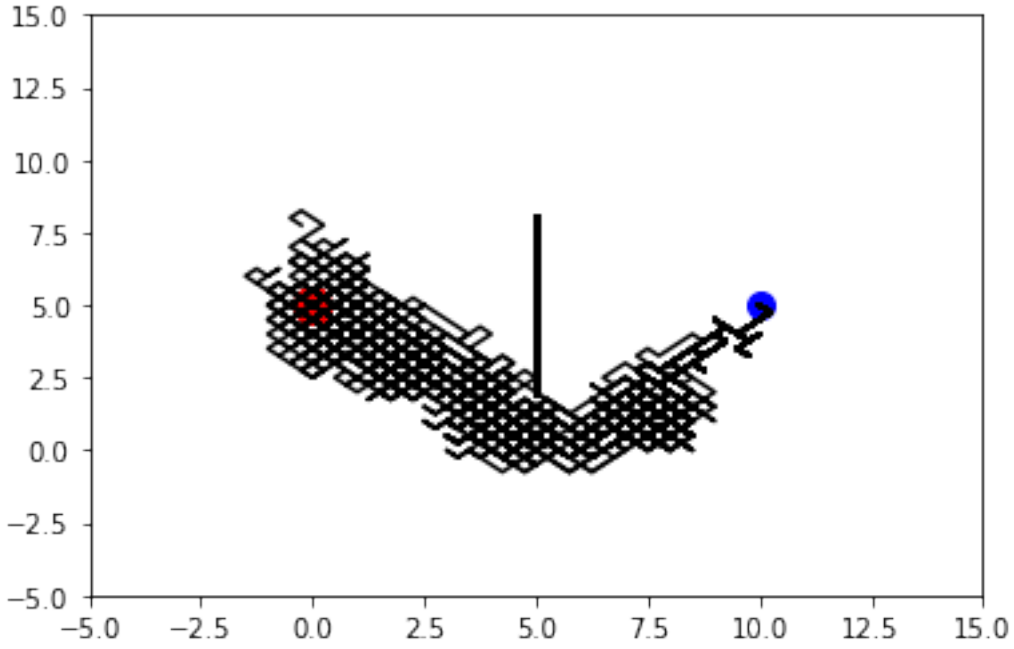

```
Ger: 64 | best fit: 0.545819714369
```



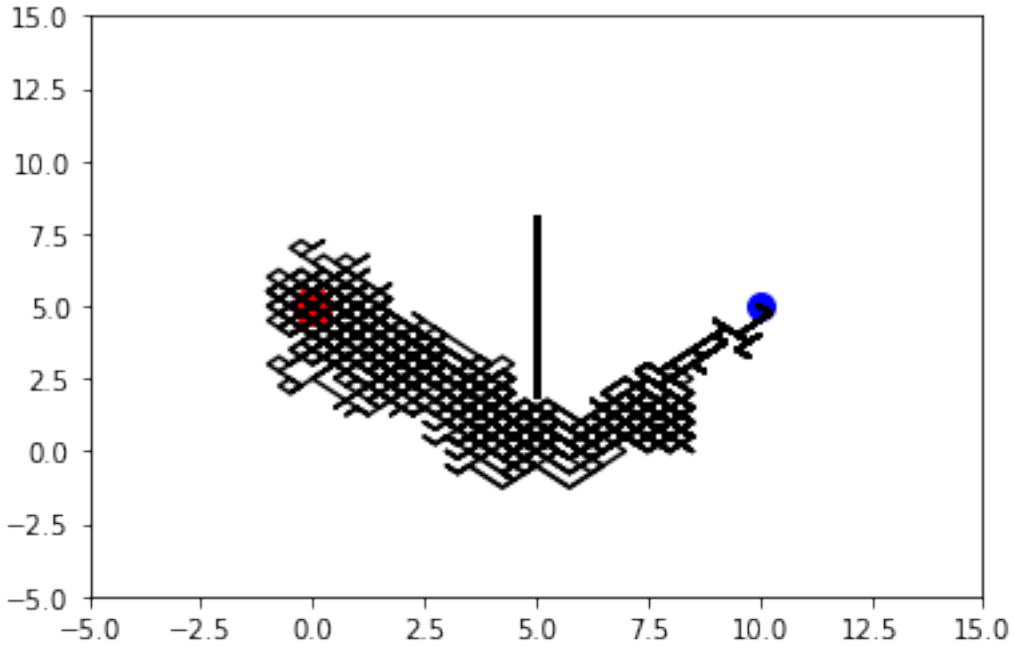
```
Ger: 65 | best fit: 0.545819714369
```



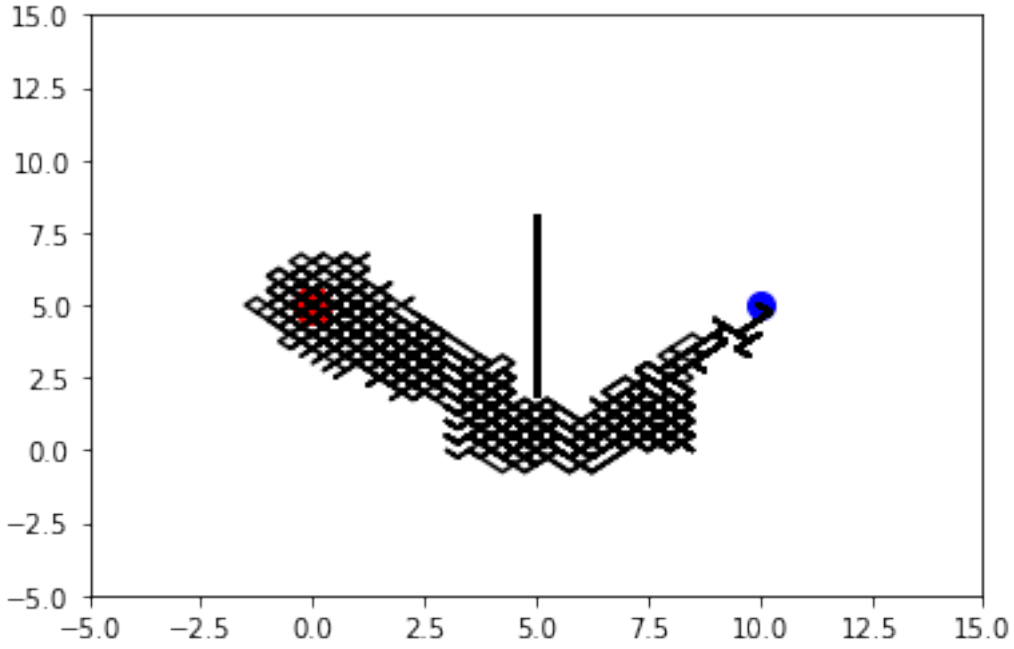

```
Ger: 66 | best fit: 0.545819714369
```



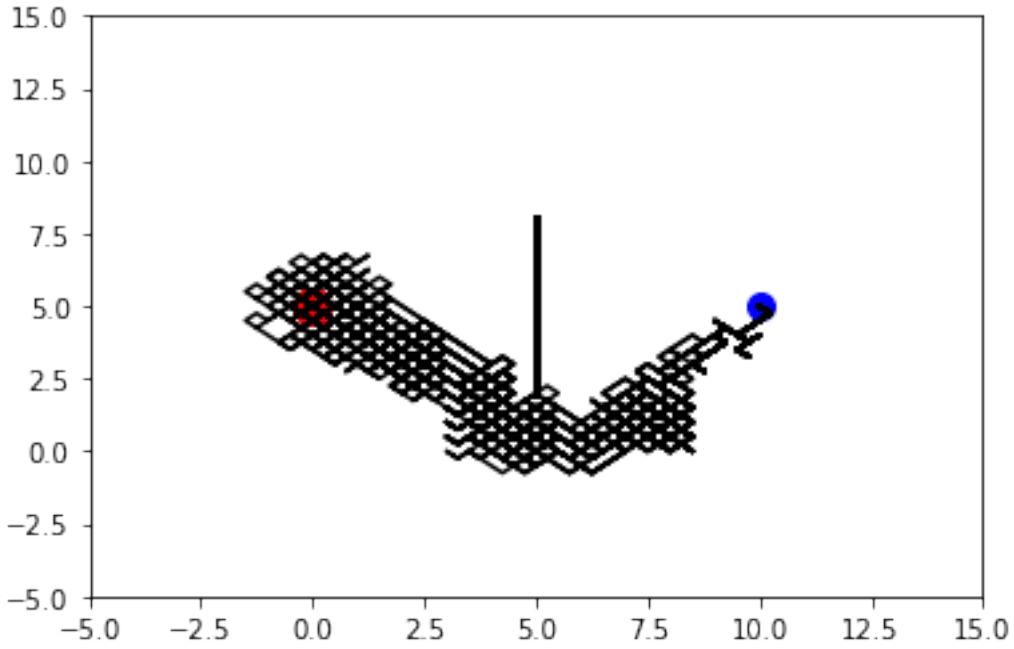
```
Ger: 67 | best fit: 0.545819714369
```



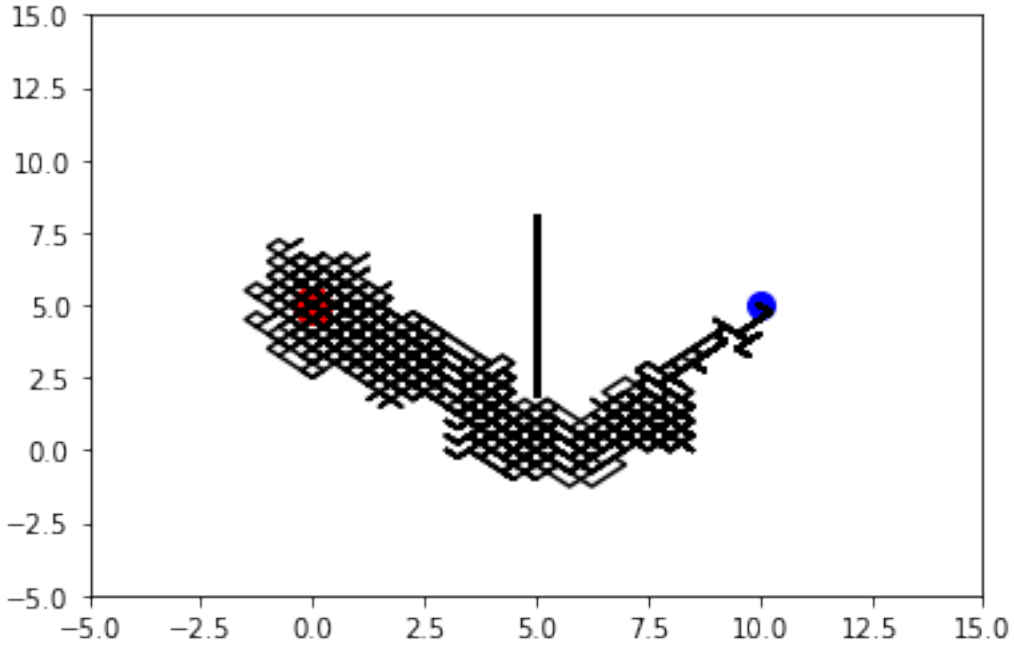

```
Ger: 68 | best fit: 0.545819714369
```



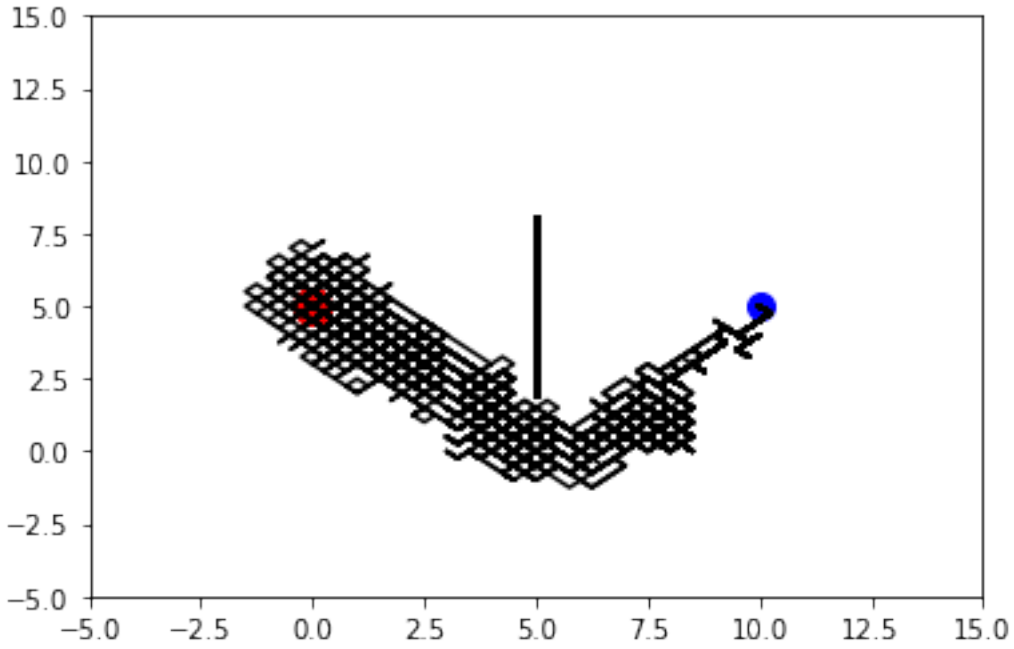
```
Ger: 69 | best fit: 0.545819714369
```



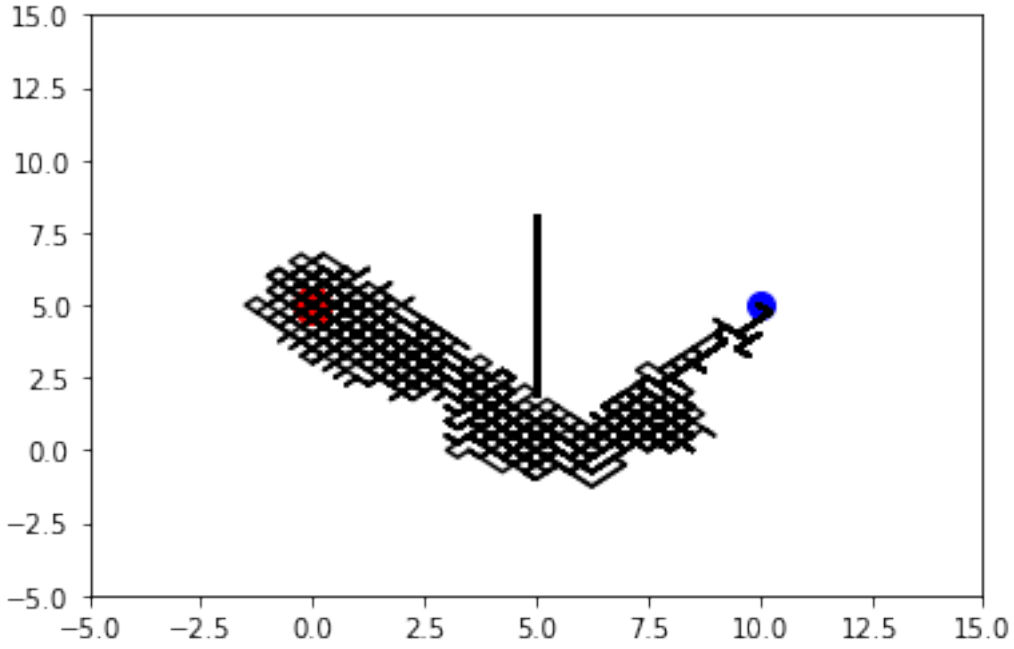

```
Ger: 70 | best fit: 0.545819714369
```



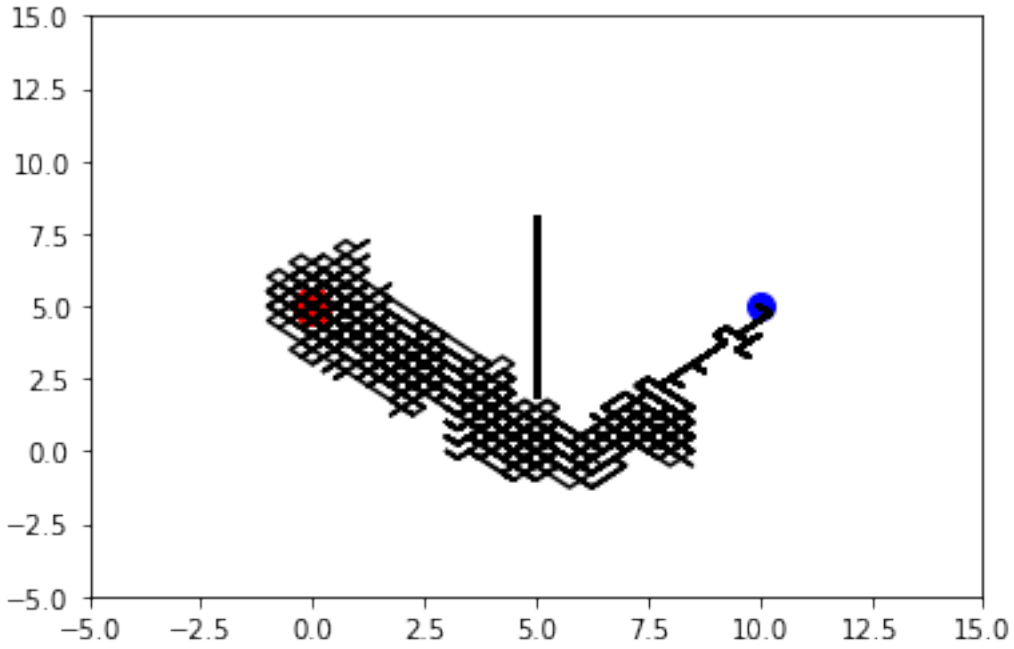
```
Ger: 71 | best fit: 0.545819714369
```



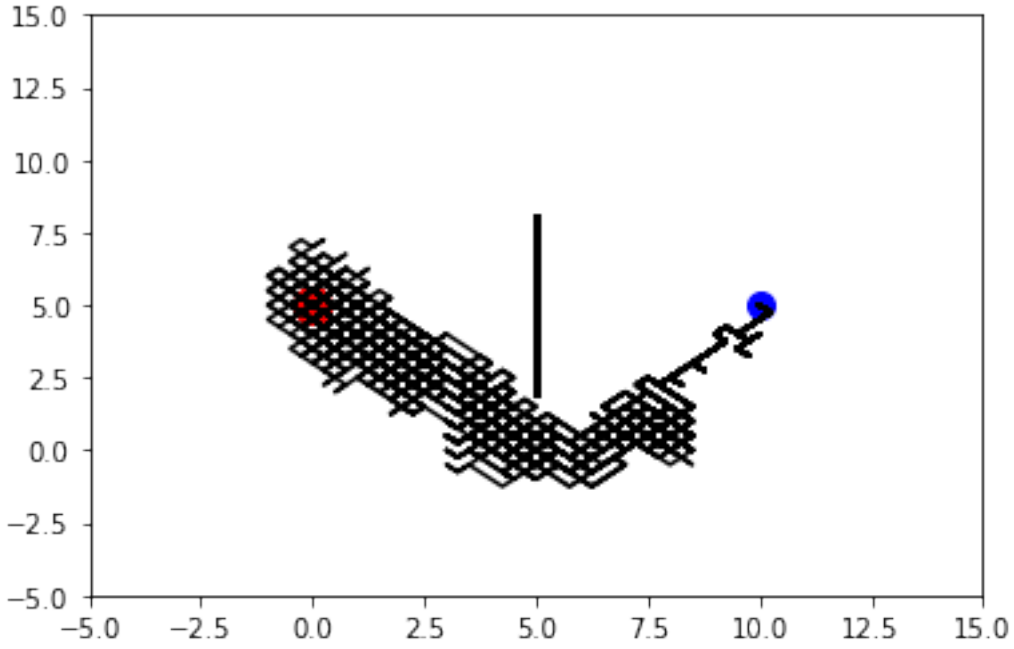

```
Ger: 72 | best fit: 0.545819714369
```



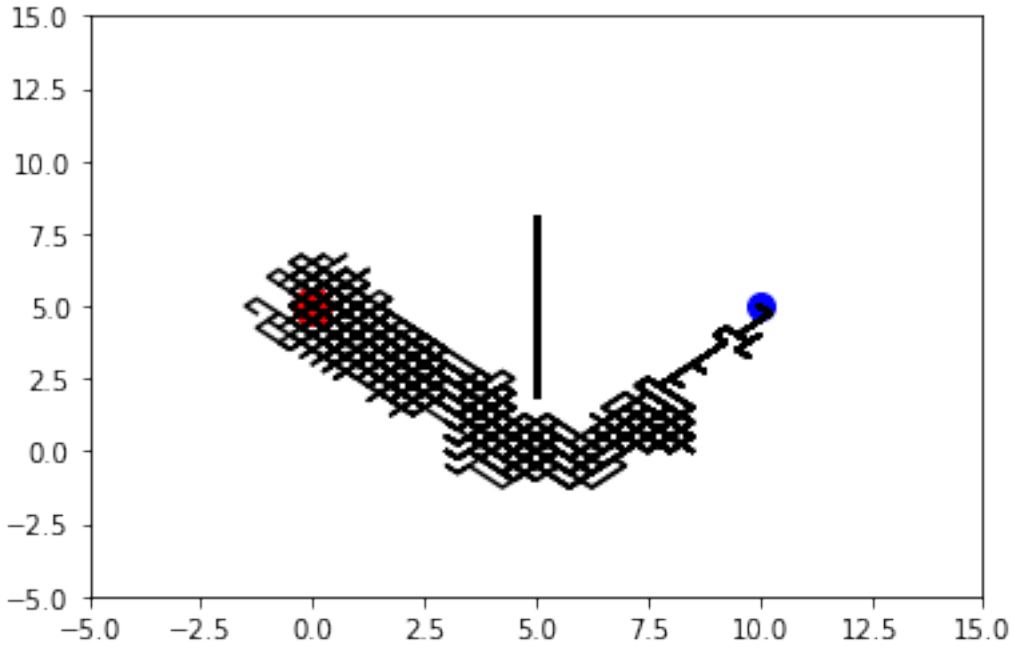
```
Ger: 73 | best fit: 0.545819714369
```



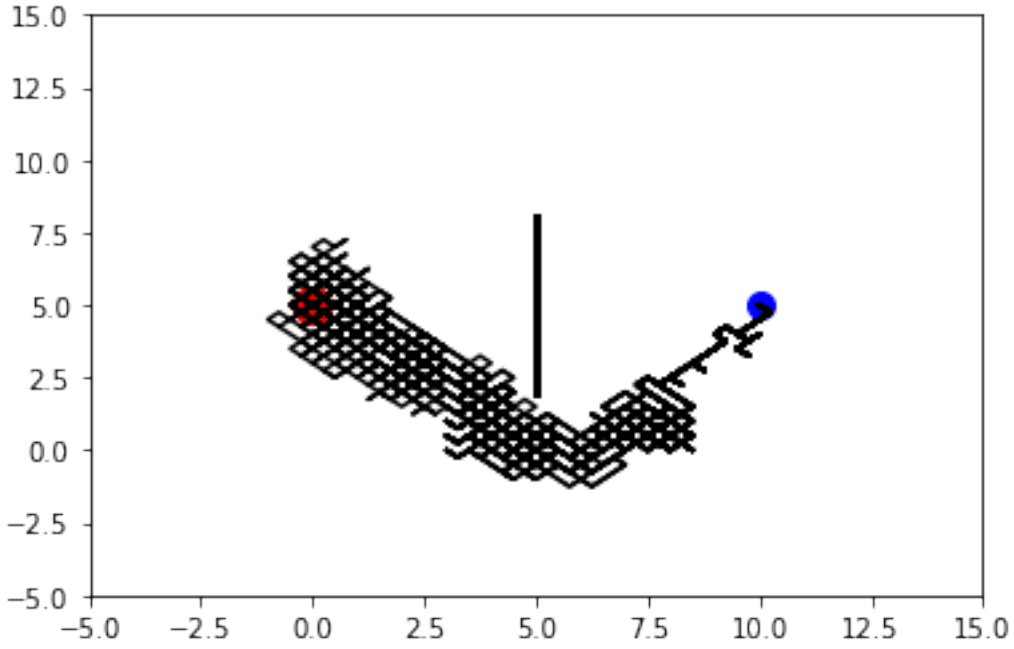

```
Ger: 74 | best fit: 0.545819714369
```



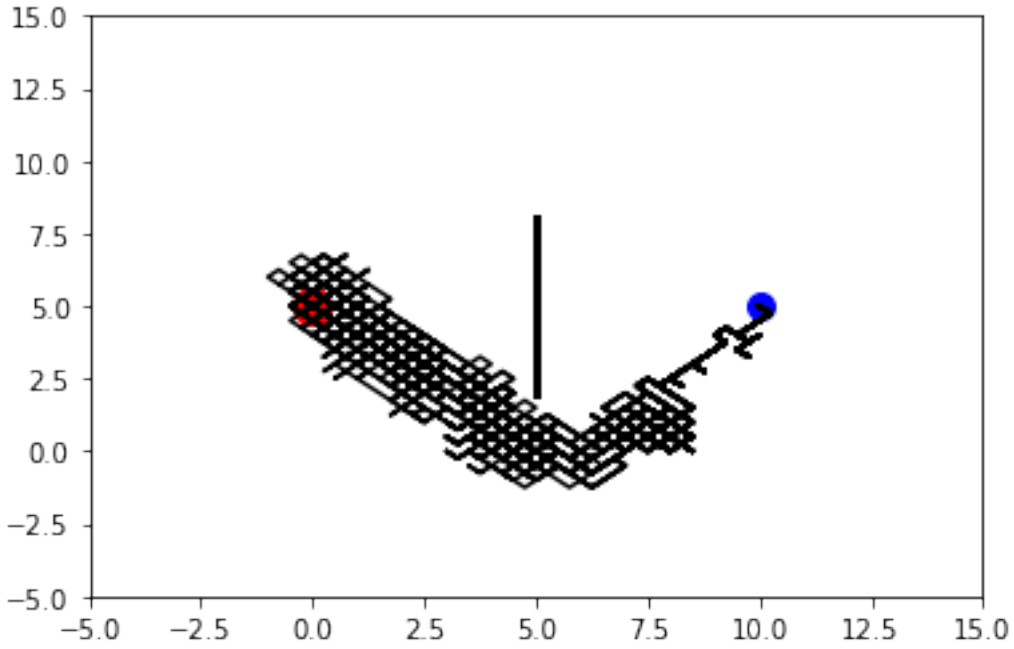
```
Ger: 75 | best fit: 0.545819714369
```



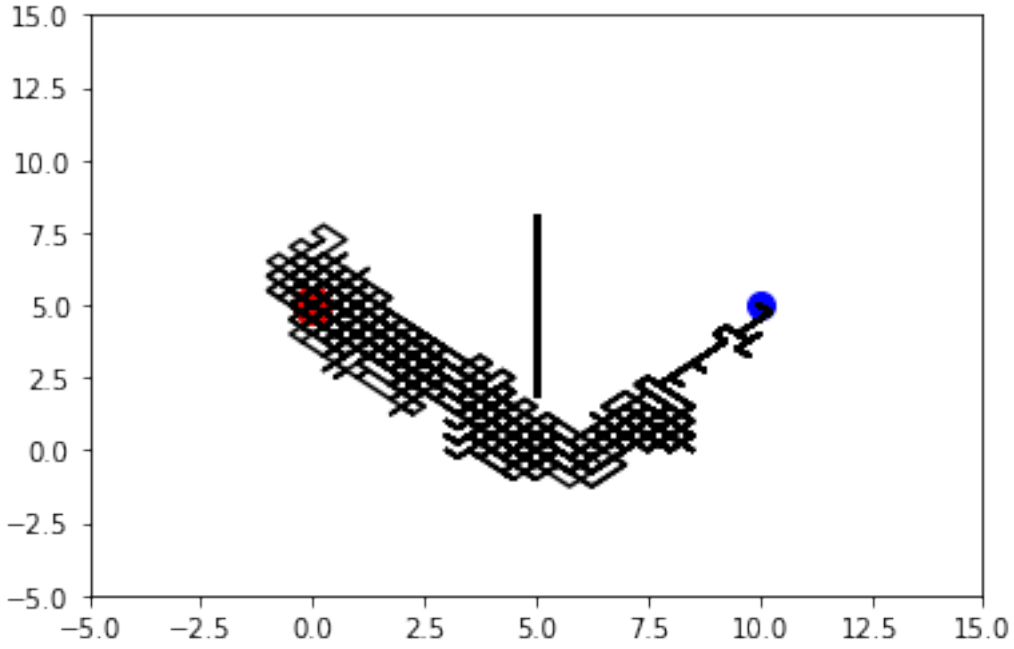

```
Ger: 76 | best fit: 0.545819714369
```



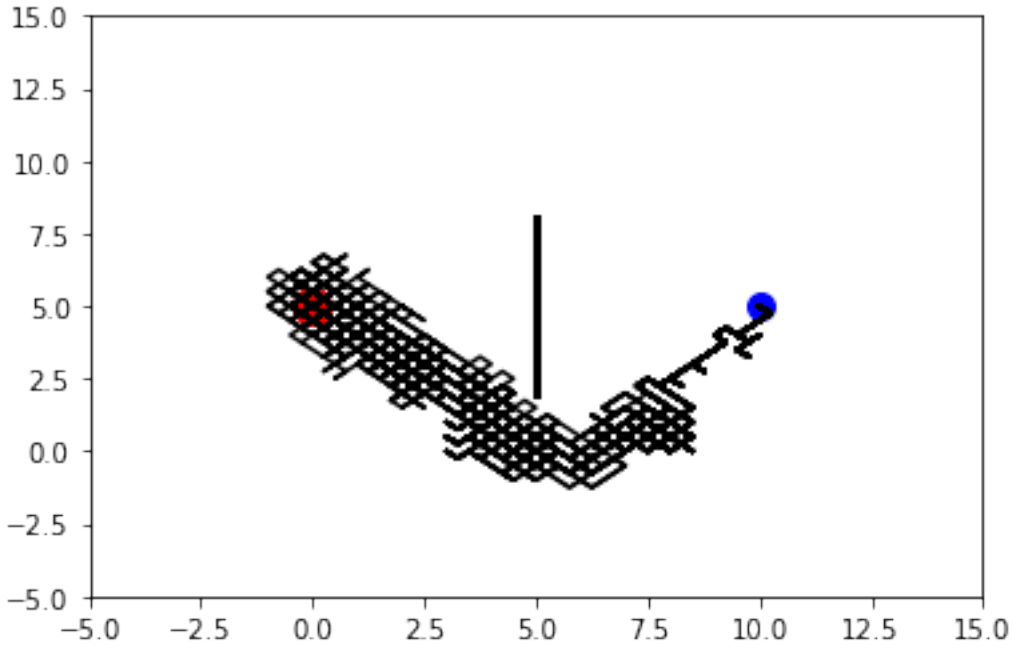
```
Ger: 77 | best fit: 0.545819714369
```



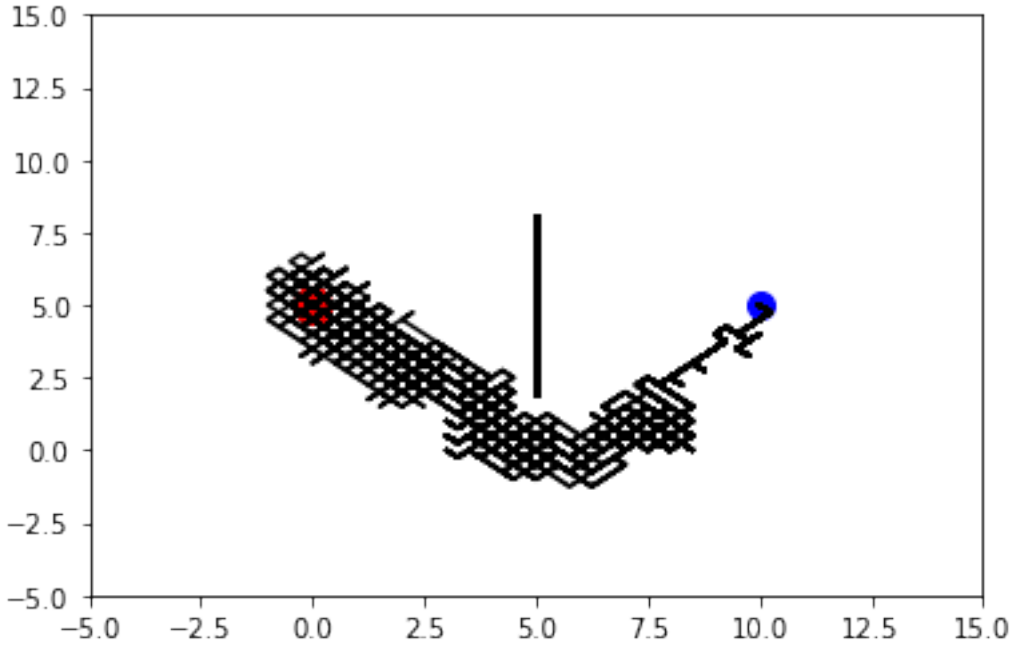

```
Ger: 78 | best fit: 0.545819714369
```



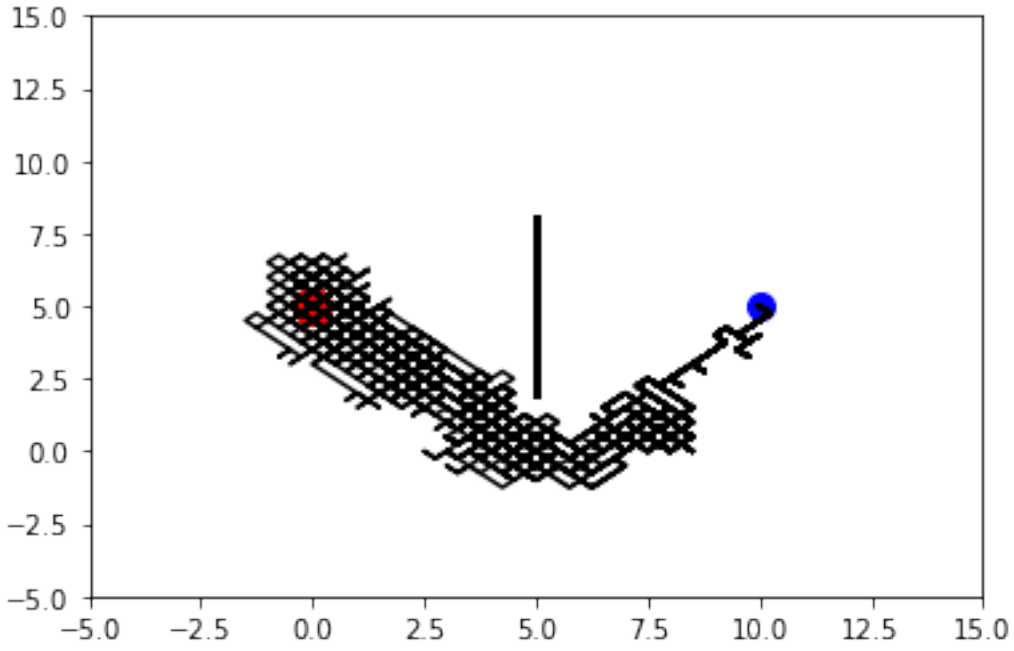
```
Ger: 79 | best fit: 0.545819714369
```



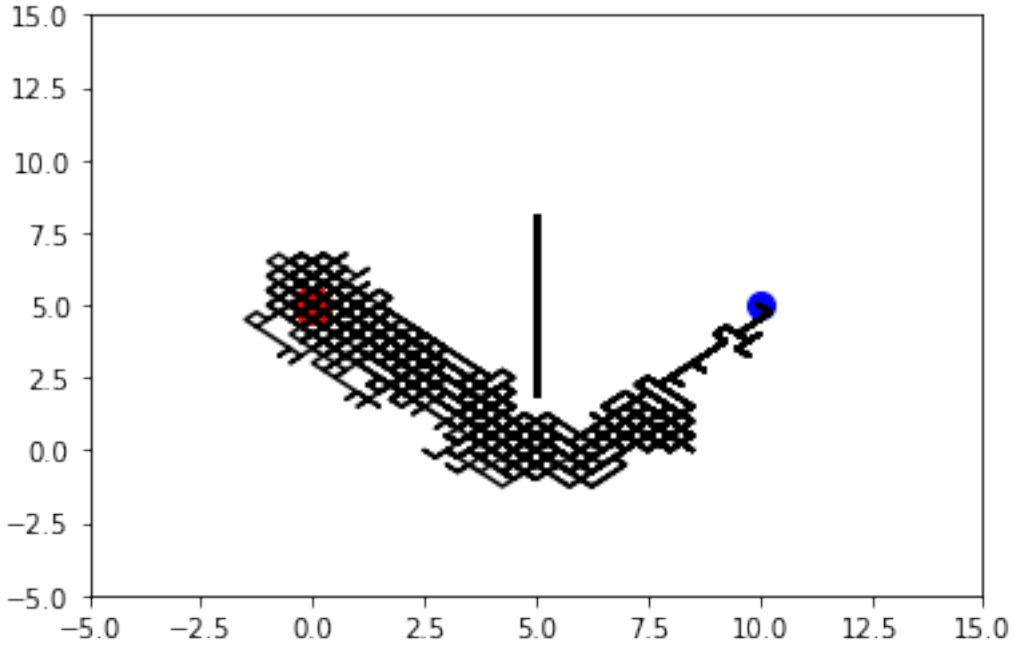

```
Ger: 80 | best fit: 0.545819714369
```



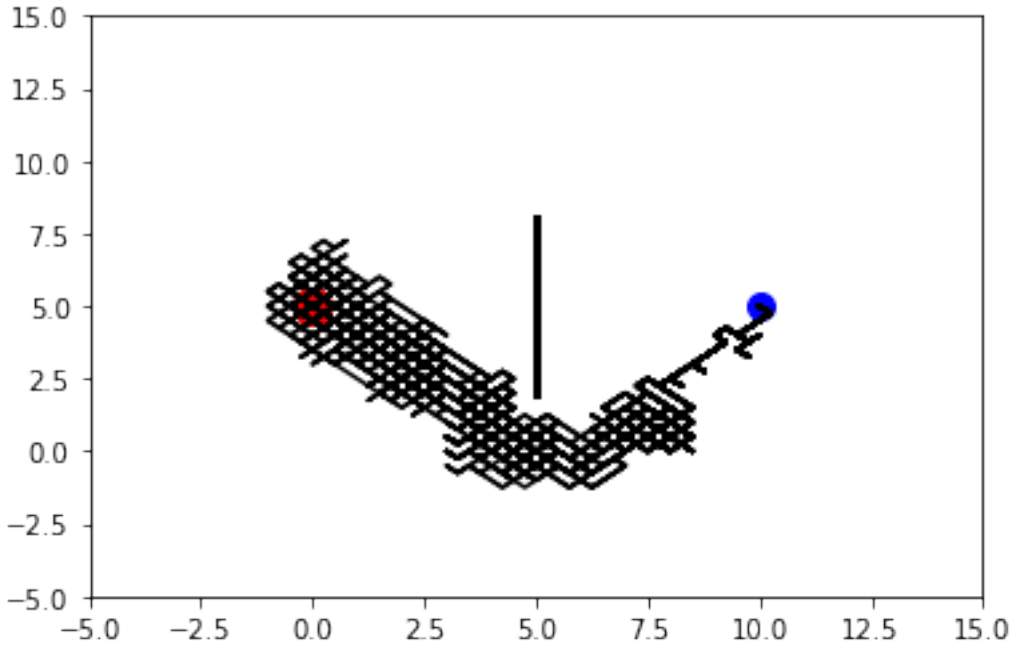
```
Ger: 81 | best fit: 0.545819714369
```



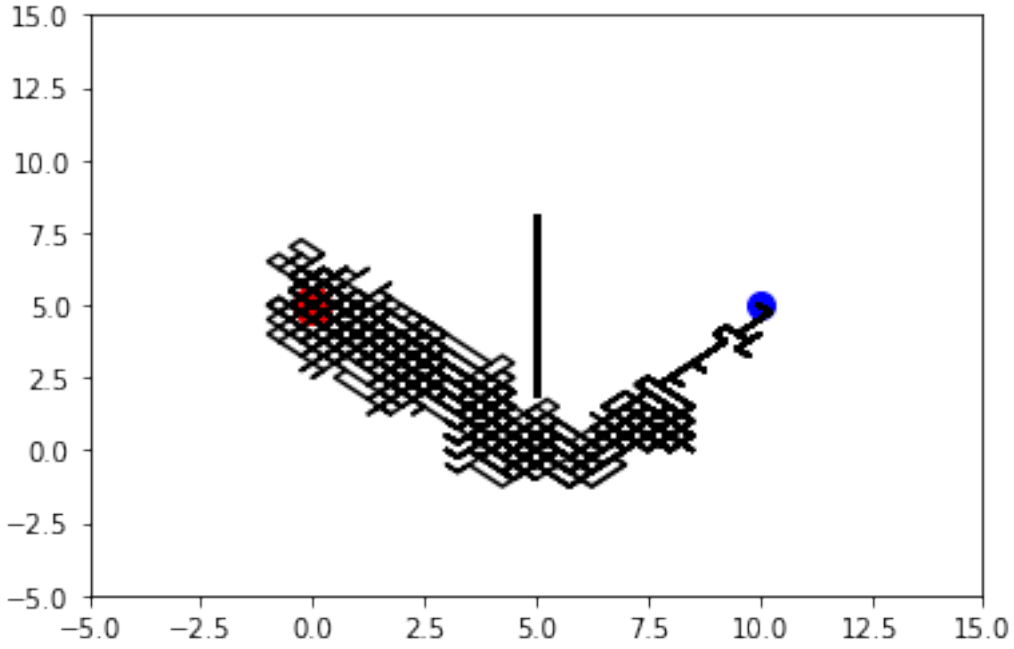

```
Ger: 82 | best fit: 0.545819714369
```



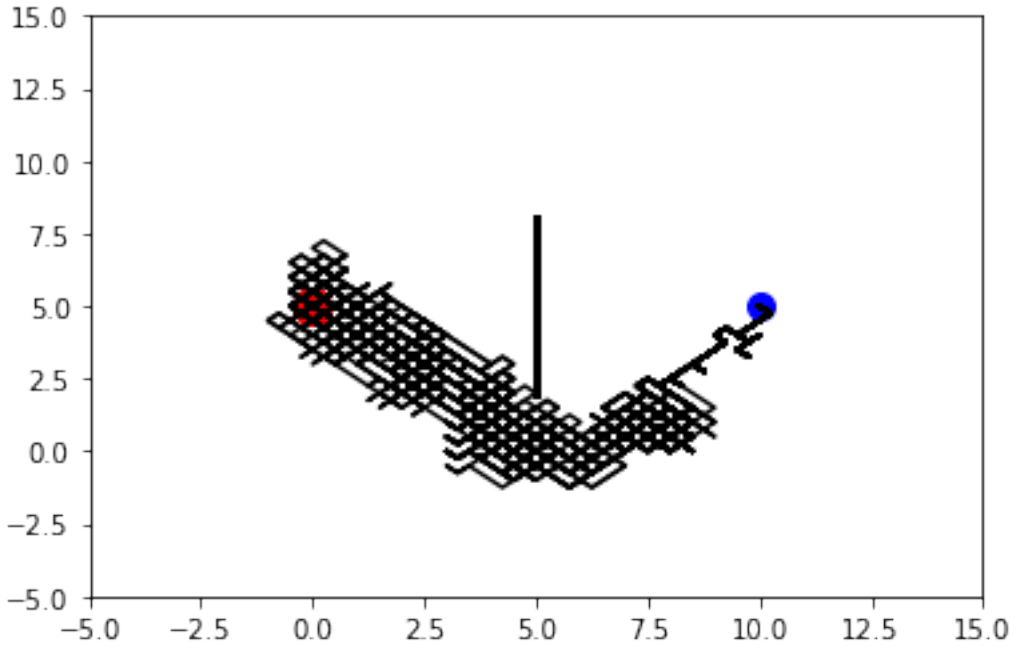
```
Ger: 83 | best fit: 0.545819714369
```



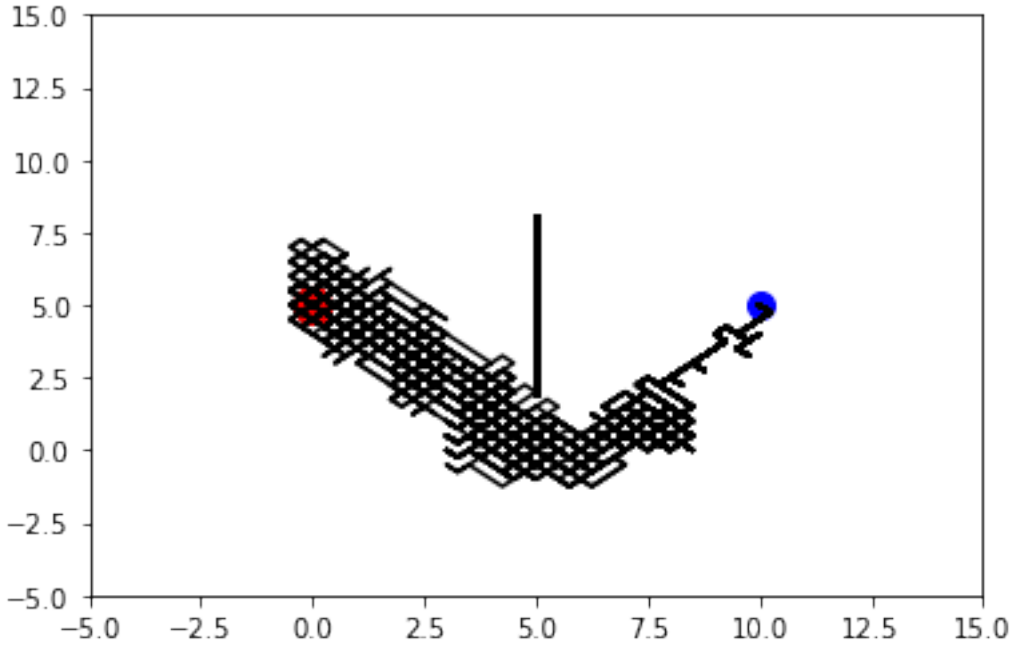

```
Ger: 84 | best fit: 0.545819714369
```



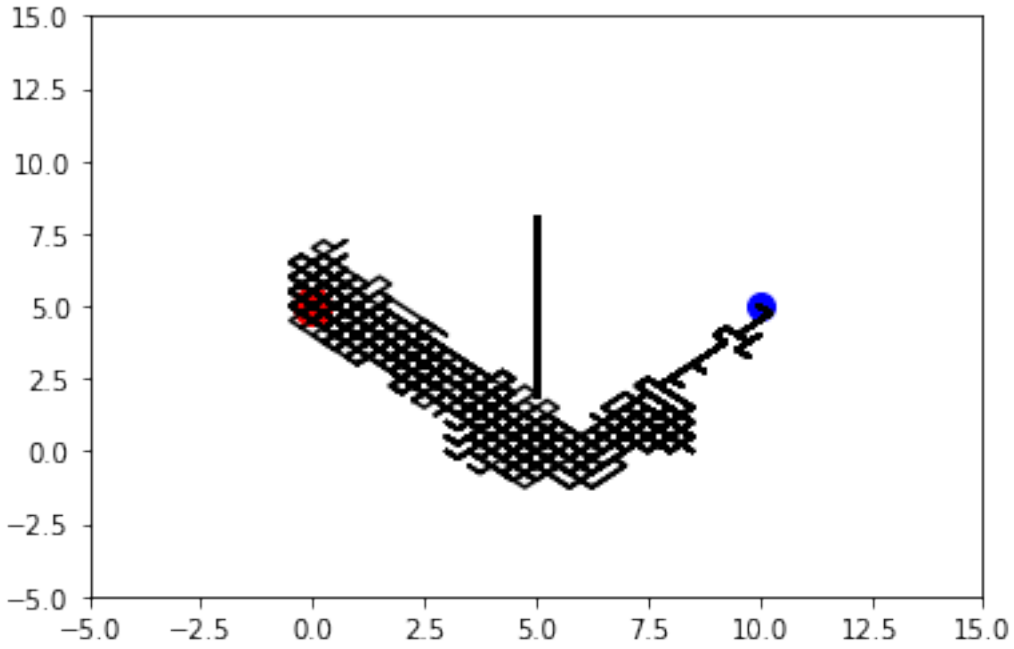
```
Ger: 85 | best fit: 0.545819714369
```



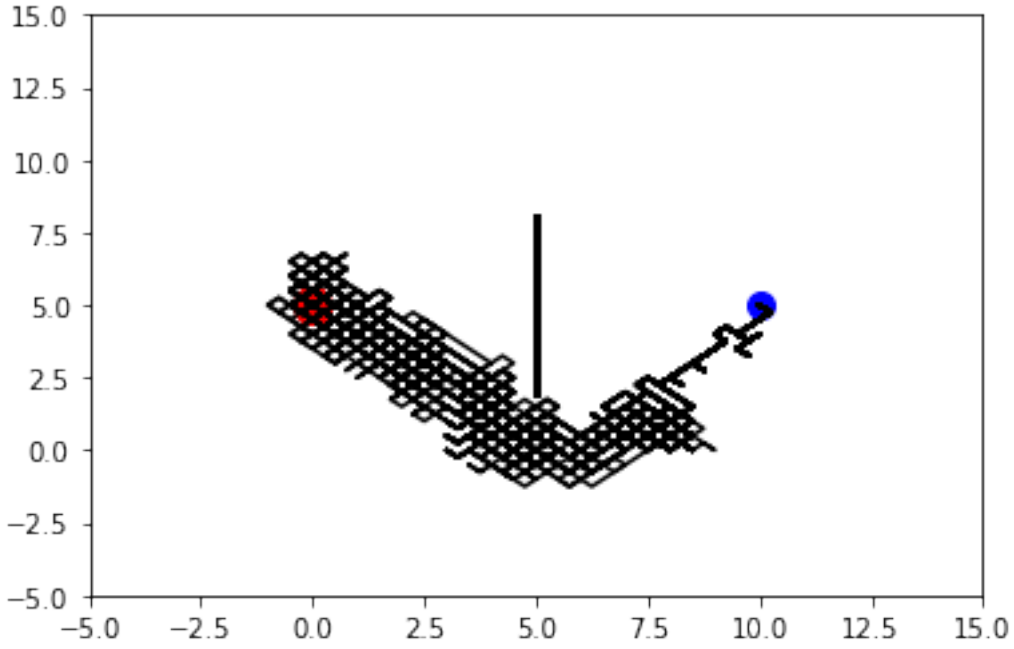

```
Ger: 86 | best fit: 0.545819714369
```



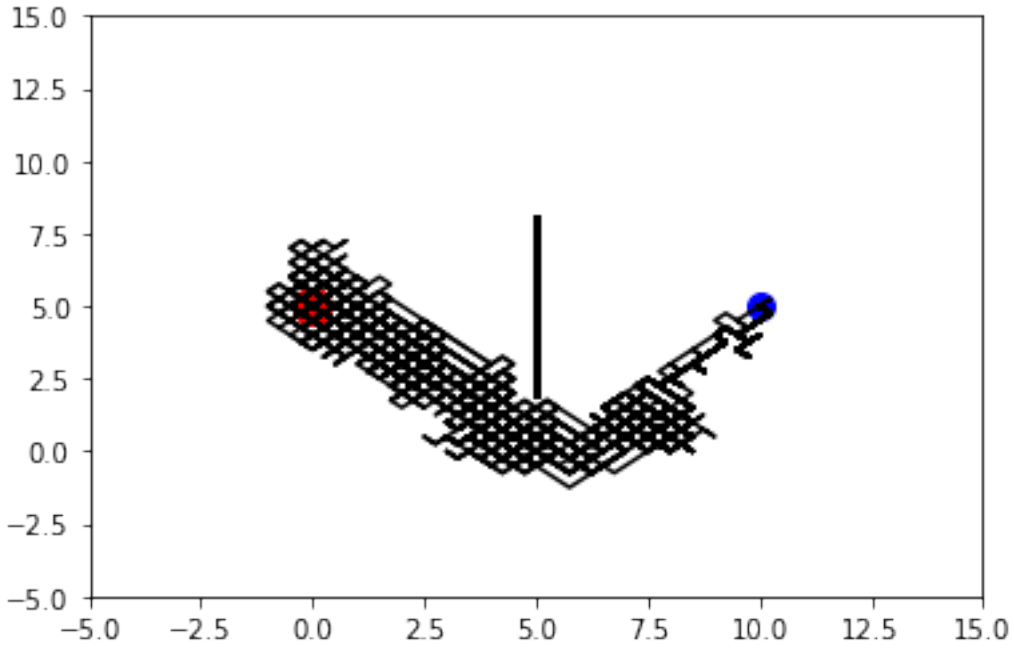
```
Ger: 87 | best fit: 0.545819714369
```



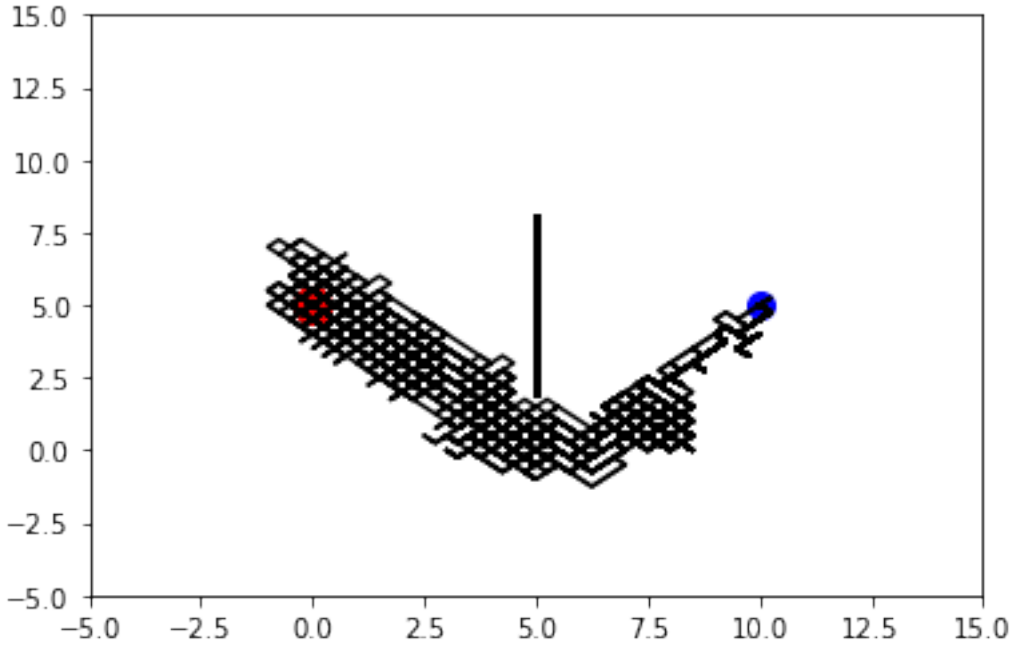

```
Ger: 88 | best fit: 0.545819714369
```



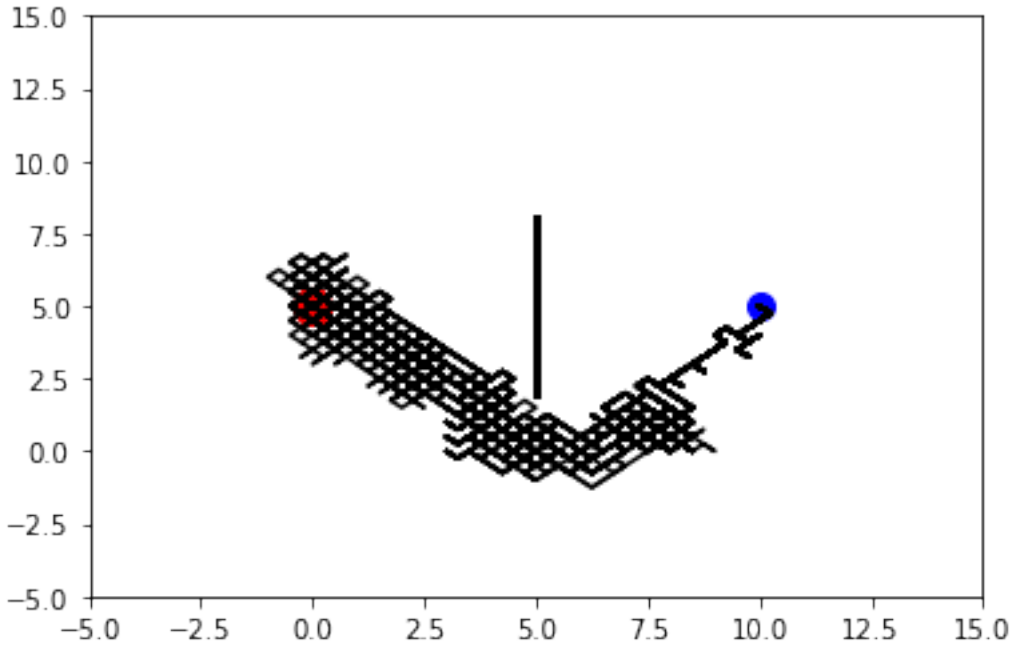
```
Ger: 89 | best fit: 0.545819714369
```



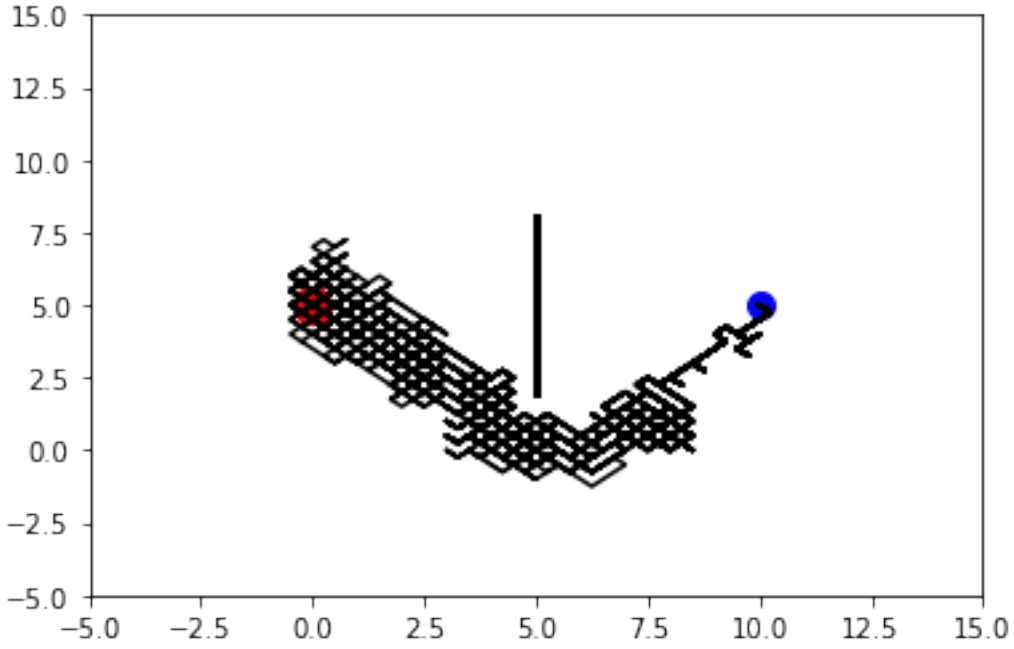

```
Ger: 90 | best fit: 0.545819714369
```



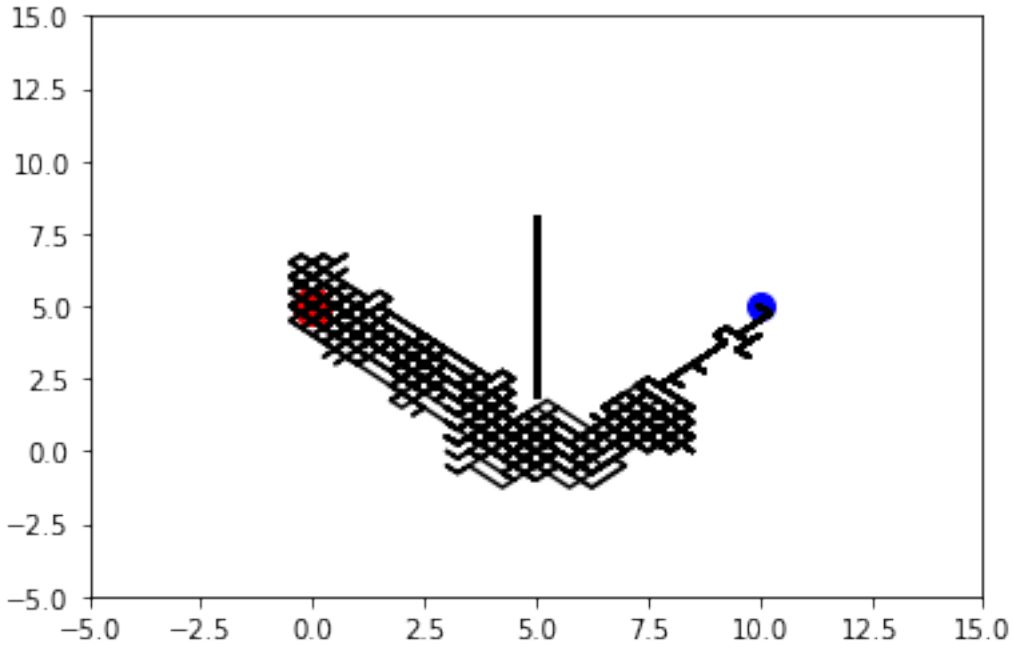
```
Ger: 91 | best fit: 0.545819714369
```



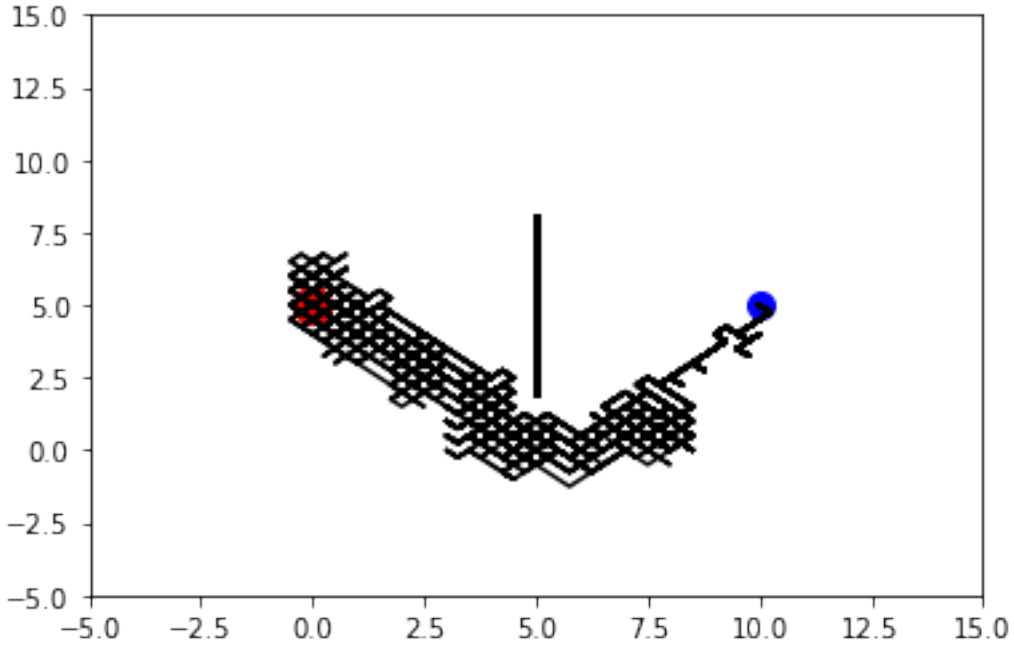

```
Ger: 92 | best fit: 0.545819714369
```



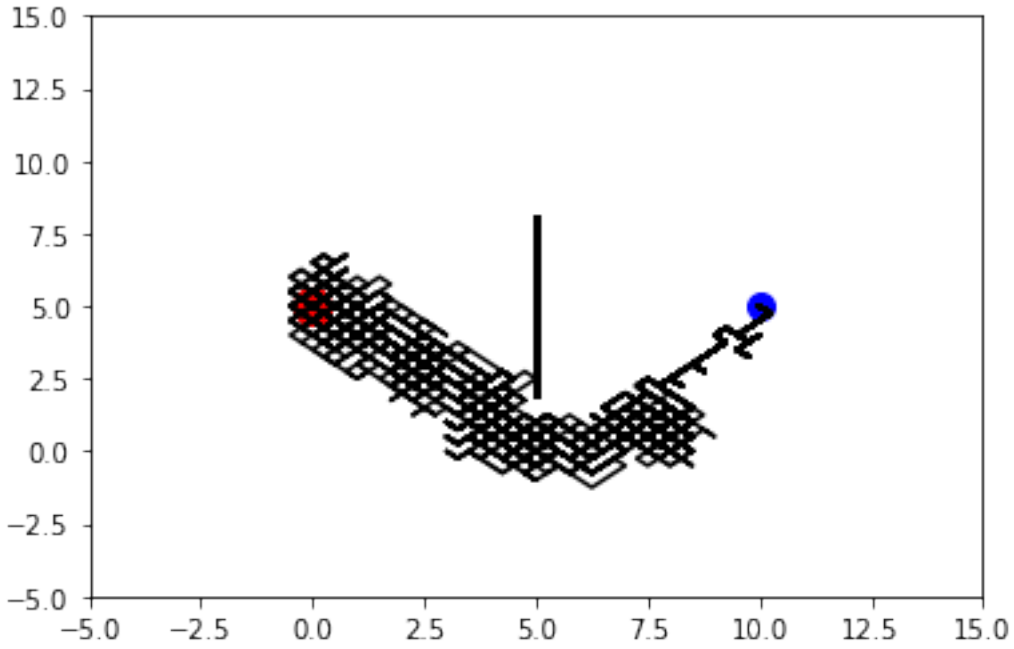
```
Ger: 93 | best fit: 0.545819714369
```



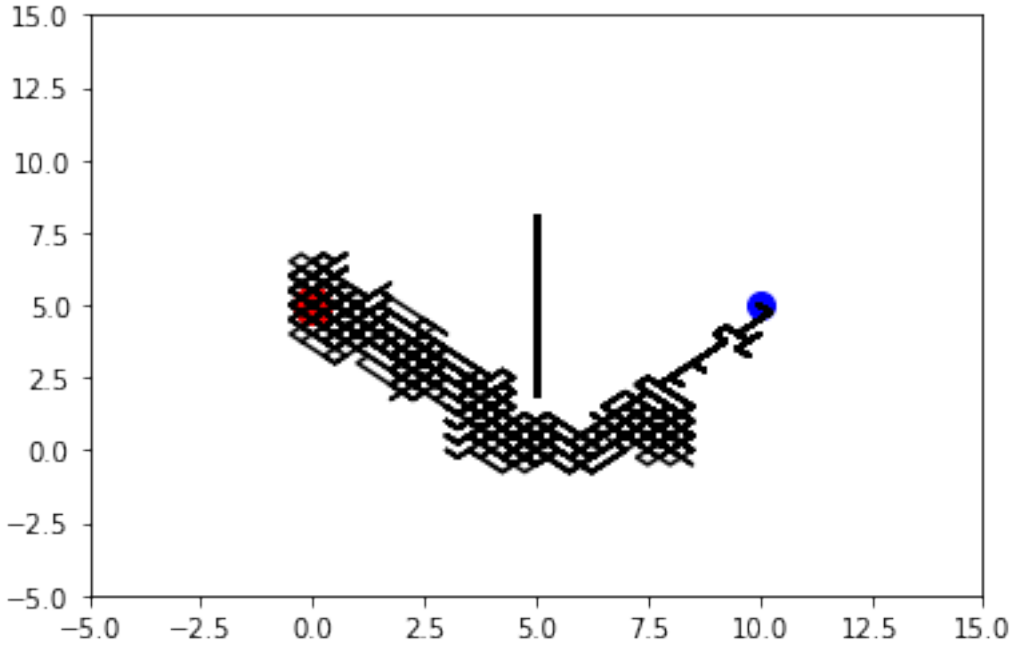

```
Ger: 94 | best fit: 0.545819714369
```



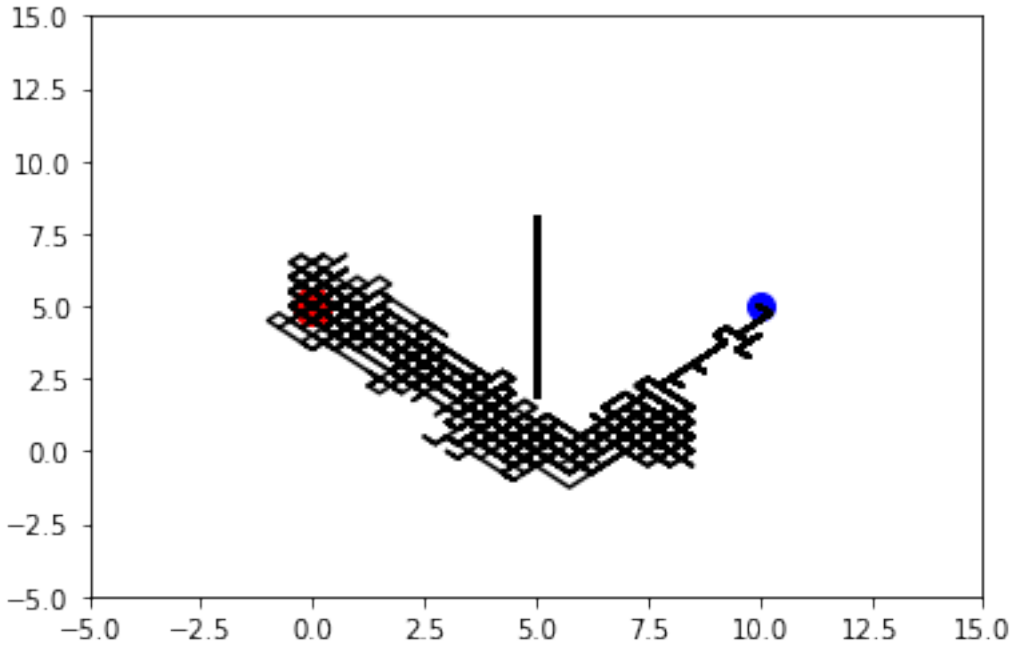
```
Ger: 95 | best fit: 0.545819714369
```



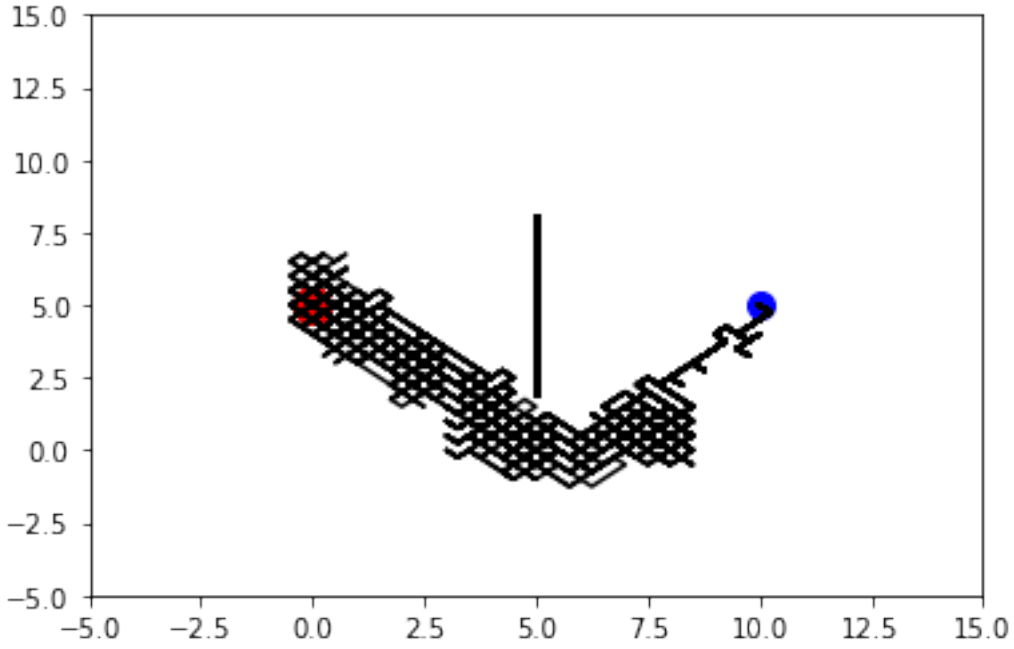

```
Ger: 96 | best fit: 0.545819714369
```



```
Ger: 97 | best fit: 0.545819714369
```




```
Ger: 98 | best fit: 0.545819714369
```



```
Ger: 99 | best fit: 0.545819714369
```

